



2 - IPv4 & IPv6

Name of report: `INR_NetEng_LAB_2_Nikita_Niakhai`



Write this commands on nodes every reboot

```
sudo dhclient -v ens3
```

`192.168.1.200` - web

`192.168.1.199` - admin

`192.168.1.198` - worker 1

Task 1 - Ports and Protocols

1. Check the open ports and listening Unix sockets against ssh (22) and http (80) on Admin and Web respectively.

Hint: use both `lsof` and `netstat`

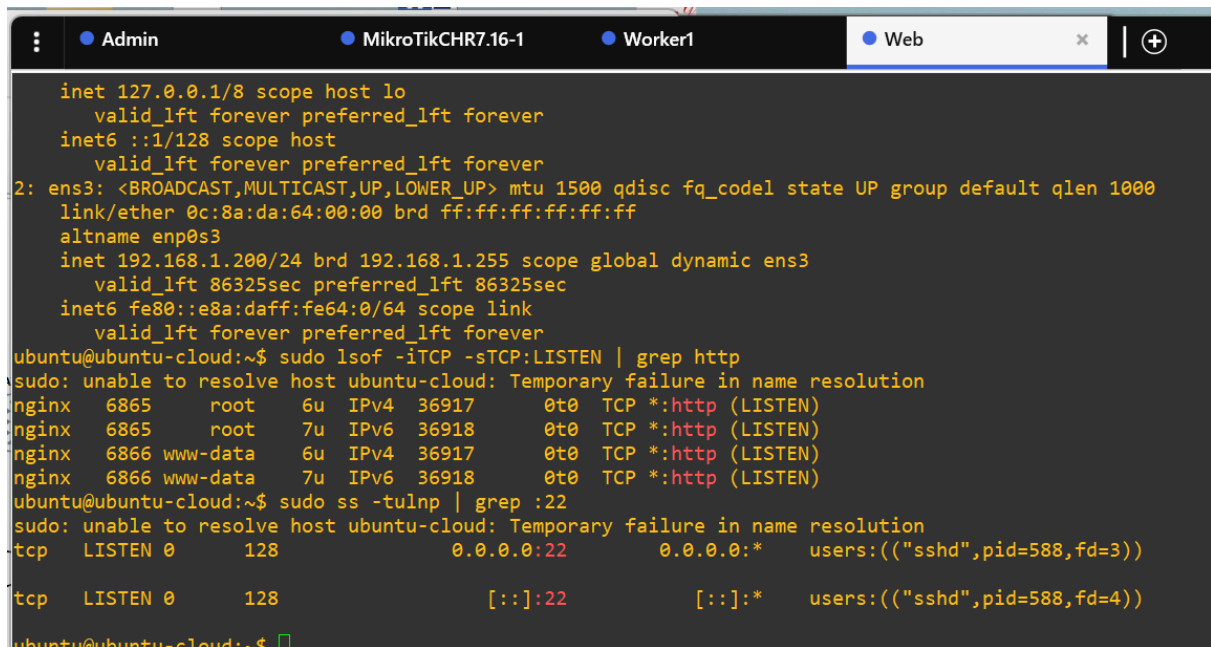


```

ubuntu@admin:~$ sudo lsof -i :22 && sudo netstat -tln | grep :22
COMMAND PID USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
sshd     620 root    3u    IPv4 17931   0t0  TCP *:ssh (LISTEN)
sshd     620 root    4u    IPv6 17942   0t0  TCP *:ssh (LISTEN)
tcp       0      0 0.0.0.0:22        0.0.0.0:*           LISTEN
tcp6      0      0 :::22           :::*             LISTEN
ubuntu@admin:~$ sudo ss -tln | grep :22
LISTEN 0      128          0.0.0.0:22        0.0.0.0:*
LISTEN 0      128          :::22            ::::*
ubuntu@admin:~$

```

Figure 2. Check for port 22 on Admin using `lsof`



```

inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 0c:8a:da:64:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 192.168.1.200/24 brd 192.168.1.255 scope global dynamic ens3
        valid_lft 86325sec preferred_lft 86325sec
    inet6 fe80::e8a:daff:fe64:0/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$ sudo lsof -iTCP -sTCP:LISTEN | grep http
sudo: unable to resolve host ubuntu-cloud: Temporary failure in name resolution
nginx    6865    root    6u    IPv4 36917   0t0  TCP *:http (LISTEN)
nginx    6865    root    7u    IPv6 36918   0t0  TCP *:http (LISTEN)
nginx    6866 www-data 6u    IPv4 36917   0t0  TCP *:http (LISTEN)
nginx    6866 www-data 7u    IPv6 36918   0t0  TCP *:http (LISTEN)
ubuntu@ubuntu-cloud:~$ sudo ss -tulnp | grep :22
sudo: unable to resolve host ubuntu-cloud: Temporary failure in name resolution
tcp      LISTEN 0      128          0.0.0.0:22        0.0.0.0:*        users:((("sshd",pid=588,fd=3))
tcp      LISTEN 0      128          :::22            ::::*            users:((("sshd",pid=588,fd=4))
ubuntu@ubuntu-cloud:~$

```

Figure 3. Check for port 80 on Web using `netstat`, `lsof`, `ss`

```

Admin MikroTikCHR7.16-1 Worker1 Web

MMM      MMM      KKK      TTTTTTTTTT      KKK
MMMM     MMMM     KKK      TTTTTTTTTT      KKK
MMM MMMM MMM III KKK KKK RRRRRR 000000 TTT III KKK KKK
MMM MM  MMM III KKKKK RRR RRR 000 000 TTT III KKKKK
MMM     MMM III KKK KKK RRRRRR 000 000 TTT III KKK KKK
MMM     MMM III KKK KKK RRR RRR 000000 TTT III KKK KKK

MikroTik RouterOS 7.16 (c) 1999-2024 https://www.mikrotik.com/

Press F1 for help

(12 messages not shown)
2025-11-21 13:41:44 system,error,critical router was rebooted without proper shutdown
2025-11-21 14:36:37 system,error,critical router was rebooted without proper shutdown
2025-11-22 16:00:42 system,error,critical router was rebooted without proper shutdown
2025-11-22 16:00:42 system,error,critical kernel failure in previous boot
2025-11-22 16:07:26 system,error,critical router was rebooted without proper shutdown
2025-11-22 16:13:06 system,error,critical login failure for user admin via local
2025-11-22 16:34:30 system,error,critical router was rebooted without proper shutdown
2025-11-22 16:48:33 system,error,critical router was rebooted without proper shutdown
[admin@MikroTik] > ip firewall nat print
Flags: X - disabled, I - invalid; D - dynamic
0 X chain=srcnat action=masquerade out-interface=ether3 ipsec-policy=out,none

1 chain=srcnat action=masquerade out-interface=ether3

2 chain=srcnat action=masquerade src-address=192.168.1.0/24 out-interface=ether3

3 chain=srcnat action=masquerade src-address=192.168.1.0/24 out-interface=ether3

4 chain=srcnat action=masquerade src-address=192.168.1.0/24 out-interface=ether3

5 chain=srcnat action=masquerade src-address=192.168.1.0/24 out-interface=ether3

6 chain=srcnat action=masquerade src-address=192.168.1.0/24 out-interface=ether3
[admin@MikroTik] > 
```

Figure 4. NAT options observed on virtual gateway

2. Scan your gateway from the outside. What are the known open ports?

Hint: use `nmap`

- `T4` option speeds up scan, `-p-` enables scanning all ports(1-65535), `-n` disables DNS lookup.

```
gns3@gns3vm: ~  
gns3@gns3vm:~$ nmap -T4 -p- -n 192.168.122.1  
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-22 17:17 U  
Nmap scan report for 192.168.122.1  
Host is up (0.000097s latency).  
Not shown: 65522 closed ports  
PORT      STATE SERVICE  
22/tcp    open  ssh  
53/tcp    open  domain  
80/tcp    open  http  
5006/tcp  open  wsm-server  
5008/tcp  open  synapsis-edge  
5010/tcp  open  telepathstart  
5015/tcp  open  fmwp  
33041/tcp open  unknown  
34289/tcp open  unknown  
36327/tcp open  unknown  
36807/tcp open  unknown  
38725/tcp open  unknown  
42459/tcp open  unknown  
  
Nmap done: 1 IP address (1 host up) scanned in 2.17 seconds  
gns3@gns3vm:~$
```

Figure 5. Scan of gateway form outside (gns3vm)

So we observe that open ports are well known: 21-ftp, 22-ssh, 23-telnet, 80-http.

3. A gateway has to be transparent, you should not see any port that is not specifically forwarded. Adjust your firewall rules to make this happen. Disable any unnecessary services and scan again.

To do that I have determined what services are unnecessary, then used command

```
/ip service disable name_of_service,...
```

The results are shown below, figure 6.

```

bad Command Name Service (line 1 column 5)
[admin@MikroTik] > /ip service print
Flags: X - DISABLED, I - INVALID
Columns: NAME, PORT, CERTIFICATE, VRF, MAX-SESSIONS
# NAME PORT CERTIFICATE VRF MAX-SESSIONS
0 X telnet 23 main 20
1 X ftp 21 20
2 X www 80 main 20
3 X ssh 22 main 20
4 X www-ssl 443 none main 20
5 X api 8728 main 20
6 X winbox 8291 main 20
7 X api-ssl 8729 none main 20
[admin@MikroTik] >

```

Figure 6. List of running services (disabled)

Because I have firewall set and real attacker don't care about 2000th port that possessed by bandwidth server I decided not to disable it. But instead I assumed that nodes coming from LAN bridge should get access to 2 services ssh and winbox, so I opened them only for LAN net, figure 7.

```

[admin@MikroTik] > /ip service set ssh address=192.168.1.0/24
[admin@MikroTik] > /ip service set winbox address=192.168.1.0/24
[admin@MikroTik] >

```

Figure 7. Enabling must-have services for LAN net

```

Admin x MikroTikC Web Worker1
ubuntu@admin:~$ nmap -T4 -p- -n 192.168.1.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-22 20:53 UTC
Nmap scan report for 192.168.1.1
Host is up (0.039s latency).
Not shown: 65533 closed ports
PORT      STATE SERVICE
53/tcp    open  domain
2000/tcp   open  cisco-sccp

Nmap scan report for 192.168.1.199
Host is up (0.0086s latency).
Not shown: 65534 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap scan report for 192.168.1.200
Host is up (0.0050s latency).
Not shown: 65533 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 256 IP addresses (3 hosts up) scanned in 36.03 seconds

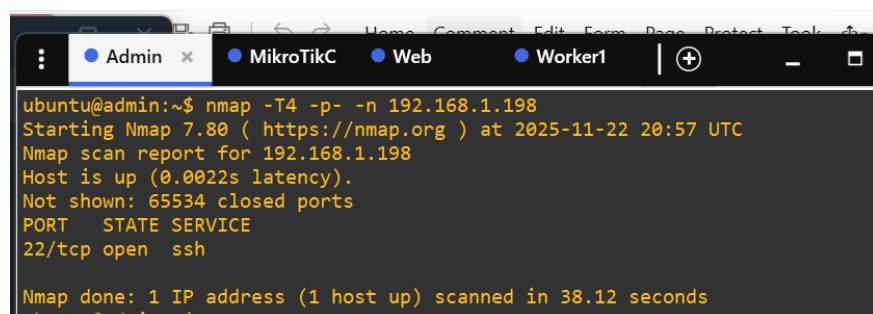
```

Figure 8. Scan again, unnecessary services are disabled.

4. It suppose that some scanners start by scanning the known ports and pinging a host to see if it is alive.

4.1. Scan the Worker VM from Admin. Can you see any ports

Results, figure 9: tcp port (22) is open for ssh

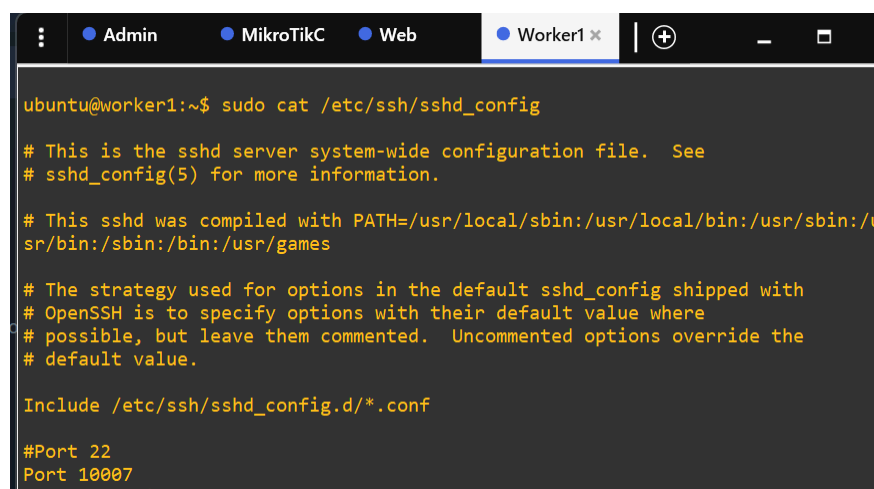


```
ubuntu@admin:~$ nmap -T4 -p- -n 192.168.1.198
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-22 20:57 UTC
Nmap scan report for 192.168.1.198
Host is up (0.0022s latency).
Not shown: 65534 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 38.12 seconds
ubuntu@admin:~$
```

Figure 9. Scan of **Worker** from **Admin**

4.2. Block ICMP traffic on Worker and change the port for SSH to one that is above 10000.



```
ubuntu@worker1:~$ sudo cat /etc/ssh/sshd_config
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/u
sr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
Port 10007
#AddressFamily any
```

Figure 10. content of the file responsible for configuration of `ssh` service

`/etc/ssh/sshd_config`

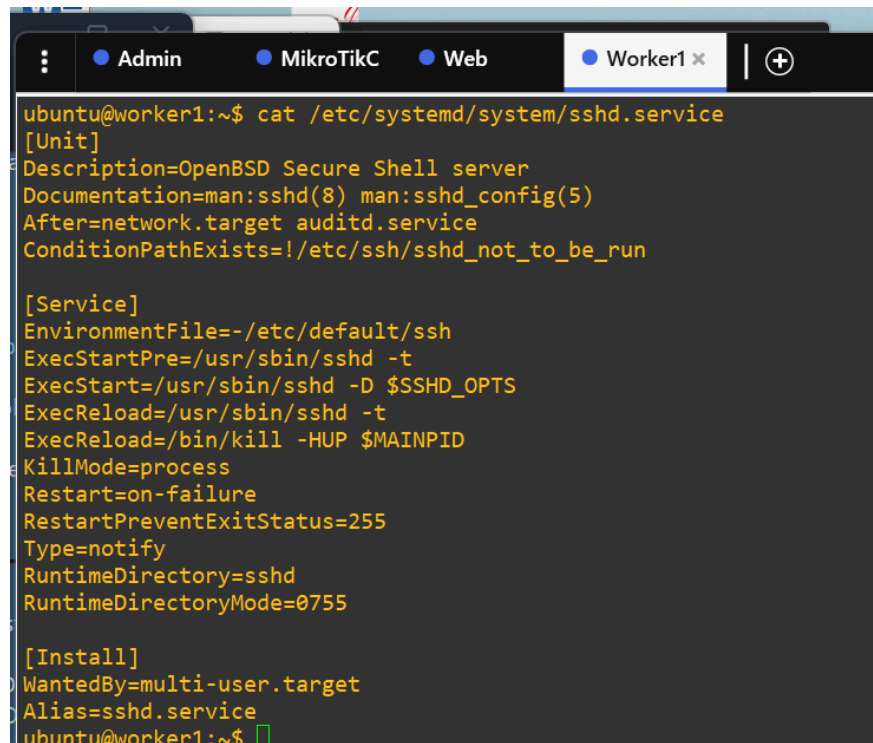
Then I restarted the service and check the status along with active port on Worker machine, figure 11.

```
ubuntu@worker1:~$ systemctl status sshd
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset:
   Active: active (running) since Sat 2025-11-22 21:01:50 UTC; 5s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Process: 900 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 901 (sshd)
      Tasks: 1 (limit: 1102)
     Memory: 1.7M
        CPU: 26ms
    CGroup: /system.slice/ssh.service
            └─901 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 22 21:01:50 worker1 systemd[1]: Starting OpenBSD Secure Shell server...
Nov 22 21:01:50 worker1 sshd[901]: Server listening on 0.0.0.0 port 10007.
Nov 22 21:01:50 worker1 sshd[901]: Server listening on :: port 10007.
Nov 22 21:01:50 worker1 systemd[1]: Started OpenBSD Secure Shell server.
lines 1-17/17 (END)
```

Figure 11. Status of ssh service inside **Worker** machine

I don't have sockets enabled, so sshd.service points to previous configuration, figure 12.

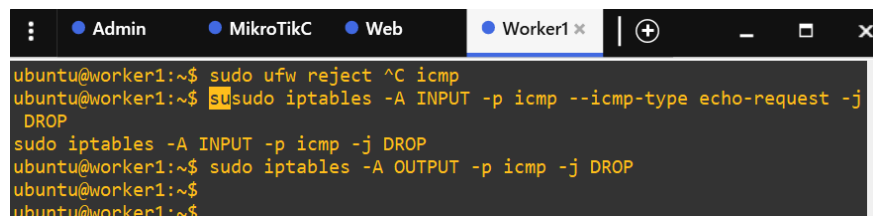


```
ubuntu@worker1:~$ cat /etc/systemd/system/ssh.service
[Unit]
Description=OpenBSD Secure Shell server
Documentation=man:sshd(8) man:sshd_config(5)
After=network.target auditd.service
ConditionPathExists=!/etc/ssh/ssh_not_to_be_run

[Service]
EnvironmentFile=-/etc/default/ssh
ExecStartPre=/usr/sbin/sshd -t
ExecStart=/usr/sbin/sshd -D $SSHD_OPTS
ExecReload=/usr/sbin/sshd -t
ExecReload=/bin/kill -HUP $MAINPID
KillMode=process
Restart=on-failure
RestartPreventExitStatus=255
Type=notify
RuntimeDirectory=ssh
RuntimeDirectoryMode=0755

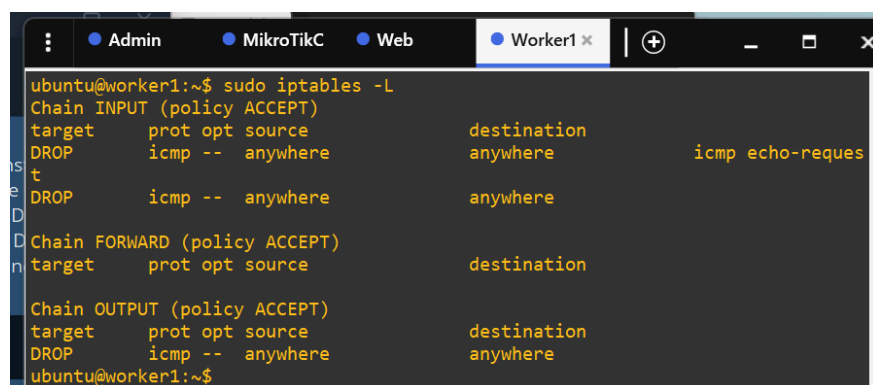
[Install]
WantedBy=multi-user.target
Alias=ssh.service
ubuntu@worker1:~$
```

Figure 12. Status of ssh service inside **Worker** machine (2)



```
ubuntu@worker1:~$ sudo ufw reject ^C icmp
ubuntu@worker1:~$ sudo iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
ubuntu@worker1:~$ sudo iptables -A INPUT -p icmp -j DROP
ubuntu@worker1:~$ sudo iptables -A OUTPUT -p icmp -j DROP
ubuntu@worker1:~$
```

Figure 13. Blocking ICMP traffic on **Worker**

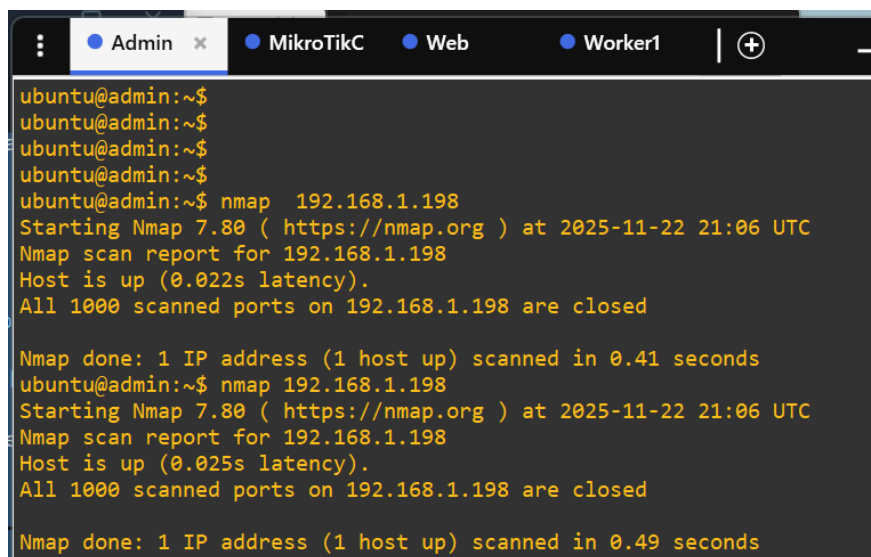


```
ubuntu@worker1:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
DROP      icmp -- anywhere          anywhere          icmp echo-reques
DROP      icmp -- anywhere          anywhere
Chain FORWARD (policy ACCEPT)
target     prot opt source                destination
Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
DROP      icmp -- anywhere          anywhere
ubuntu@worker1:~$
```

Figure 14. **IPTABLES** that show the results of new rules.

4.3. Scan it without extra arguments.

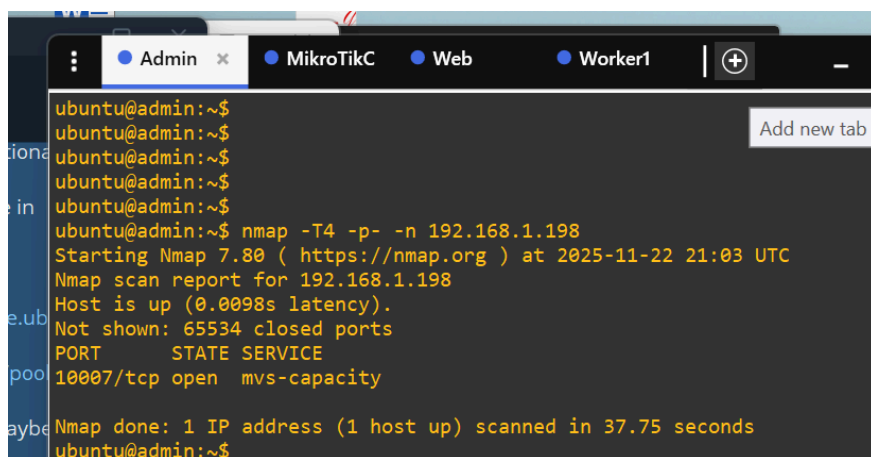
No open port were found, figure 15.



```
ubuntu@admin:~$  
ubuntu@admin:~$  
ubuntu@admin:~$  
ubuntu@admin:~$  
ubuntu@admin:~$ nmap 192.168.1.198  
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-22 21:06 UTC  
Nmap scan report for 192.168.1.198  
Host is up (0.022s latency).  
All 1000 scanned ports on 192.168.1.198 are closed  
  
Nmap done: 1 IP address (1 host up) scanned in 0.41 seconds  
ubuntu@admin:~$ nmap 192.168.1.198  
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-22 21:06 UTC  
Nmap scan report for 192.168.1.198  
Host is up (0.025s latency).  
All 1000 scanned ports on 192.168.1.198 are closed  
  
Nmap done: 1 IP address (1 host up) scanned in 0.49 seconds
```

Figure 15. Scan of **Worker** from **Admin** without extras

4.4. Now make necessary changes to the command to force the scan on all possible ports, figure 16.



```
ubuntu@admin:~$  
ubuntu@admin:~$  
ubuntu@admin:~$  
ubuntu@admin:~$  
ubuntu@admin:~$ nmap -T4 -p- -n 192.168.1.198  
Starting Nmap 7.80 ( https://nmap.org ) at 2025-11-22 21:03 UTC  
Nmap scan report for 192.168.1.198  
Host is up (0.0098s latency).  
Not shown: 65534 closed ports  
PORT      STATE SERVICE  
10007/tcp open  mvs-capacity  
  
Nmap done: 1 IP address (1 host up) scanned in 37.75 seconds  
ubuntu@admin:~$
```

Figure 16. Scan of **Worker** from **Admin** with extra arguments.

4.5. Gather some information about your open ports on Web (ssh and http).

Note. Don't paste the scan results, summarize them in the answer and include them as an appendix of your submission in Moodle.

Flags description:

Flag	What it does (short)	Why you need it for the lab
-A	Enable OS detection, version detection, script scanning, and traceroute	
--min-rate	will increase port scan	
-sS	TCP SYN scan (stealth, fast)	Default best scan
-sV	Detect service + exact version (nginx 1.18.0, OpenSSH 8.9p1, etc.)	Shows real software
-sC	Run default script engine (http-title, ssh-hostkey, http-methods, banner grabbing...)	Grabs web page title, allowed HTTP methods, SSH keys
-O	OS detection (guesses Ubuntu 22.04, kernel version)	Shows OS fingerprint
-p 22,80	Scan only these ports (or remove to scan all)	Focused on web + SSH
-v	Verbose – see what's happening live	Good for screenshots
--reason	Explain why a port is open/filtered	Proof for lab report
--open	Show only open ports	Clean output

Task 2 - Traffic Captures

In some cases, you might need to take a look at the traffic sent and received from your machines to understand what is going on. You will be sniffing the traffic of your External services. For this,

you can use Wireshark which has an integration with GNS3 or tcpdump from the machines.

1. Access your Web's http page from outside and capture the traffic between the gateway and the bridged interface.

Can you see what is being sent?

What kind of information can you get from this?

What do the headers mean?

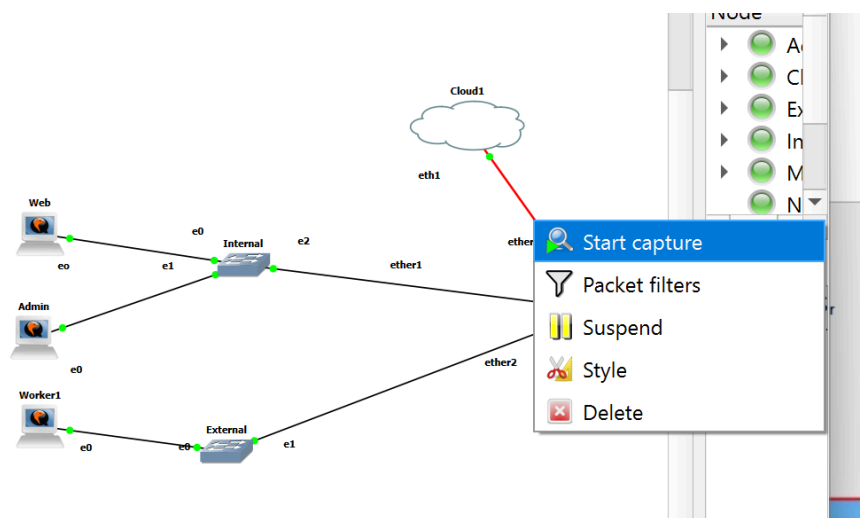


Figure 17. Starting **Wireshark** integration inside GNS3.

I started pinging google.com from Admin machine and immediately received packets in Wireshark.

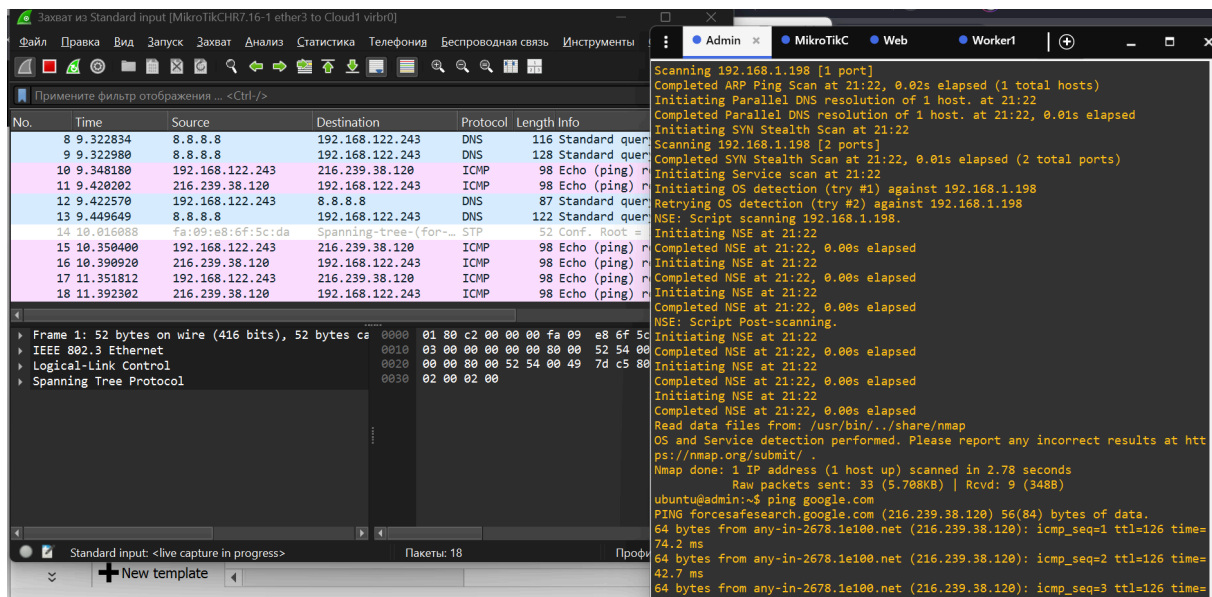


Figure 18. MNDP, CDP, LLDP, STP

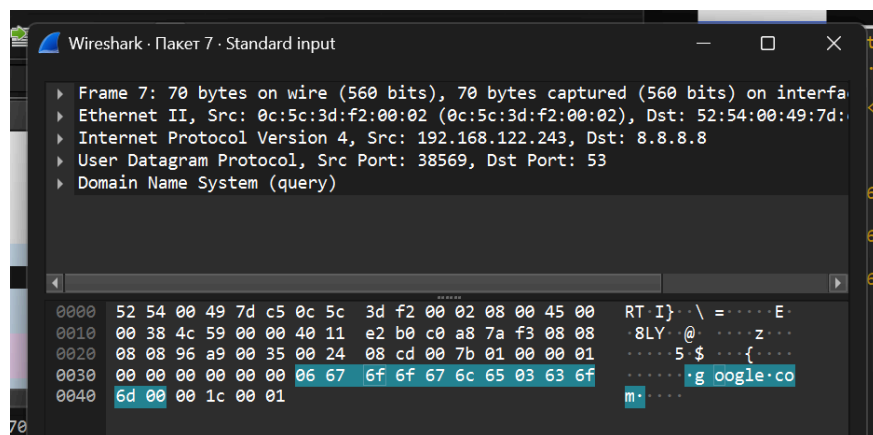


Figure 19. pinging content

No.	Time	Source	Destination	Protocol	Length	Info
11	8.557266	192.168.1.199	192.168.1.200	TCP	74	47400 → 80 [SYN] Seq=0
12	8.558102	192.168.1.200	192.168.1.199	TCP	74	80 → 47400 [SYN, ACK] Seq=1
13	8.558781	192.168.1.199	192.168.1.200	TCP	66	47400 → 80 [ACK] Seq=1
14	8.559446	192.168.1.199	192.168.1.200	HTTP	143	GET / HTTP/1.1
15	8.560631	192.168.1.200	192.168.1.199	TCP	66	80 → 47400 [ACK] Seq=1
16	8.560671	192.168.1.200	192.168.1.199	HTTP	925	HTTP/1.1 200 OK (text/html)
17	8.561242	192.168.1.199	192.168.1.200	TCP	66	47400 → 80 [ACK] Seq=78
18	8.623110	192.168.1.199	192.168.1.200	TCP	66	47400 → 80 [FIN, ACK] Seq=78
19	8.624335	192.168.1.200	192.168.1.199	TCP	66	80 → 47400 [FIN, ACK] Seq=78
20	8.624752	192.168.1.199	192.168.1.200	TCP	66	47400 → 80 [ACK] Seq=79

Figure 20. Content of accessing http web page.

```

Internet Protocol Version 4, Src: 192.168.1.200, Dst: 192.168.1.199
Transmission Control Protocol, Src Port: 80, Dst Port: 47400, Seq: 1, Ack: 78, Len: 859
Hypertext Transfer Protocol
  Line-based text data: text/html (25 lines)
    <!DOCTYPE html>\n
    <html>\n
    <head>\n
    <title>Welcome to nginx!</title>\n
    <style>\n
    body {\n
      width: 35em;\n
      margin: 0 auto;\n
      font-family: Tahoma, Verdana, Arial, sans-serif;\n
    }\n
    </style>\n
    </head>\n
    <body>\n
    <h1>Welcome to nginx!</h1>\n
    <p>If you see this page, the nginx web server is successfully installed and\n
    working. Further configuration is required.</p>\n
    \n
    <p>For online documentation and support please refer to\n
    <a href="http://nginx.org/">nginx.org</a>.<br/>\n
    Commercial support is available at\n
    <a href="http://nginx.com/">nginx.com</a>.</p>\n
    \n
    <p><em>Thank you for using nginx.</em></p>\n
    </body>\n
    </html>\n
  
```

Figure 20. Content of accessing http web page (2)

We can clearly see (figures 18-20) that because HTTP has no encryption – every request and response is sent in plaintext, so any attacker on the path can read everything.

HTTP headers are metadata fields included in both requests and responses. They carry additional information: user-agent (what browser/program is used), cookies (session data), accepted content types, etc.

Since everything is unencrypted, an attacker can read and modify all headers and the actual page content in transit.

- SSH to the Admin from outside and capture the traffic (make sure to start capturing before connecting to the server).
Can you see what is being sent?
What kind of information can you get from this?
What are the names of the ciphers used?

Mostly TCP packets are seen. Packets are encrypted, data (any plaintext) is not seen.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::ed0:edff:fe74::	ff02::1	MNDP	210	5678 → 5678 Len=148
2	0.000095	192.168.49.2	255.255.255.255	MNDP	190	5678 → 5678 Len=148
3	0.000122	0c:d0:ed:74:00:02	CDP/VTP/DTP/PagP/UD...	CDP	122	Device ID: MikroTik
4	0.000130	0c:d0:ed:74:00:02	LLDP_Multicast	LLDP	159	MA/0c:d0:ed:74:00:00
5	11.205522	192.168.49.1	192.168.49.2	TCP	74	45620 → 2222 [SYN] Se
6	11.208250	192.168.49.2	192.168.49.1	TCP	74	2222 → 45620 [SYN, AC
7	11.208435	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
8	11.208943	192.168.49.1	192.168.49.2	TCP	109	45620 → 2222 [PSH, AC
9	11.210569	192.168.49.2	192.168.49.1	TCP	66	2222 → 45620 [ACK] Se
10	11.258520	192.168.49.2	192.168.49.1	TCP	108	2222 → 45620 [PSH, AC
11	11.258596	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
12	11.259736	192.168.49.1	192.168.49.2	TCP	1514	45620 → 2222 [ACK] Se
13	11.259806	192.168.49.1	192.168.49.2	TCP	154	45620 → 2222 [PSH, AC
14	11.261130	192.168.49.2	192.168.49.1	TCP	66	2222 → 45620 [ACK] Se
15	11.261263	192.168.49.2	192.168.49.1	TCP	66	2222 → 45620 [ACK] Se
16	11.264759	192.168.49.2	192.168.49.1	TCP	1178	2222 → 45620 [PSH, AC
17	11.305959	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
18	11.364545	192.168.49.1	192.168.49.2	TCP	1274	45620 → 2222 [PSH, AC
19	11.387495	192.168.49.2	192.168.49.1	TCP	1514	2222 → 45620 [ACK] Se
20	11.387582	192.168.49.2	192.168.49.1	TCP	182	2222 → 45620 [PSH, AC
21	11.388312	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
22	11.388372	192.168.49.1	192.168.49.2	TCP	66	45620 → 2222 [ACK] Se
23	12.690901	192.168.49.1	192.168.49.2	TCP	82	45620 → 2222 [PSH, AC
24	12.734268	192.168.49.2	192.168.49.1	TCP	66	2222 → 45620 [ACK] Se

Frame 1: 210 bytes on wire (1680 bits), 210 bytes capture	0000	33 33 00 00 00 01 0c d0 ed 74 00 00
Ethernet II, Src: 0c:d0:ed:74:00:02 (0c:d0:ed:74:00:02),	0010	76 85 00 9c 11 01 fe 80 00 00 00 00
Internet Protocol Version 6, Src: fe80::ed0:edff:fe74::2,	0020	ed ff fe 74 00 02 ff 02 00 00 00 00
User Datagram Protocol, Src Port: 5678, Dst Port: 5678	0030	00 00 00 00 00 01 16 2e 16 2e 00 00
Mikrotik Neighbor Discovery Protocol	0040	00 00 00 01 00 06 0c d0 ed 74 00 00
	0050	4d 69 6b 72 6f 54 69 6b 00 07 00 20
	0060	20 28 73 74 61 62 6c 65 29 20 32 33
	0070	39 2d 32 30 20 31 33 3a 30 30 3a 33

Figure 21. Content during ssh connection.

```
SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.10
.....7T.u e.7[Q;...].1sntrup761x25519-sha512@openssh.com,curve25519-sha256,curve25519-sha256@lib
ssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha25
6,diffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sha256,ext-info
-c,kex-strict-c-v00@openssh.com...ssh-ed25519-cert-v01@openssh.com,ecdsa-sha2-nistp256-cert-v01@ope
nssh.com,ecdsa-sha2-nistp384-cert-v01@openssh.com,ecdsa-sha2-nistp521-cert-v01@openssh.com,sk-ssh-ed
25519-cert-v01@openssh.com,sk-ecdsa-sha2-nistp256-cert-v01@openssh.com,rsa-sha2-512-cert-v01@openssh
.com,rsa-sha2-256-cert-v01@openssh.com,ssh-ed25519,ecdsa-sha2-nistp256,ecdsa-sha2-nistp384,ecdsa-sha
2-nistp521,sk-ssh-ed25519@openssh.com,sk-ecdsa-sha2-nistp256@openssh.com,rsa-sha2-512,rsa-sha2-256...
.lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@o
penssh.com...lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,
aes256-gcm@openssh.com...umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh
.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.co
m,hmac-sha2-256,hmac-sha2-512,hmac-sha1...umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha
2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,um
ac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-sha1...none,zlib@openssh.com,zlib...none,zlib@
openssh.com,zlib...
...T
....E.L.Qy9....f...&curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nis
tp384,ecdh-sha2-nistp521,sntrup761x25519-sha512@openssh.com,diffie-hellman-group-exchange-sha256,dif
fie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sha256,kex-strict-s-
v00@openssh.com...9rsa-sha2-512,rsa-sha2-256,ecdsa-sha2-nistp256,ssh-ed25519...lchacha20-poly1305@op
enssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com...lchacha20
-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com
...umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm
@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-s
ha2-512,hmac-sha1...umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,
hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,hma
c-sha2-256,hmac-sha2-512,hmac-sha1...none,zlib@openssh.com...none,zlib@openssh.com...
.....
.....EEN'.b..R..1...N...X.#.].yU...qT...k...JD[.f..1.....JH.....>R.(..j).2g...1.aLR.Zw.1
...M...[...e...4P...^.....<Nc.vim.....S.Y.t.mp.<.....[.<.:c.:I....)I...D...*.m..{O'.b.BJ.Ea..
0.;R.....U?{8/Xs.....T...=>...x1..V...zcD..3=.....RD.|..o...w...78V2.?....&.tS...NS...
d%....8...=|.b...4D....&....>:p9....U...R.....H...Cj9.I...[.....$H.&L..y..S../..D
"U...[n...Z.....65o.....sq..l.*...6.-u|.8,....z.z.Z...H
```

Figure 22. Capturing TCP flow whilst connected via ssh

Results of the capture, figure 22: algorithms are matched, SSH version exchange, supported enc. algorithms.

3. Configure the Burp suite as a proxy on your machine and intercept your HTTP traffic.

Show that you can modify the contents by changing something in the request.

Why are you able to do this here and not in an SSH connection?

- 1 - configure nginx proxy on Ubuntu server for 8080th port.
- 2 - connect Web's server using Burp
- 3 - see results (enjoy the capturing process)

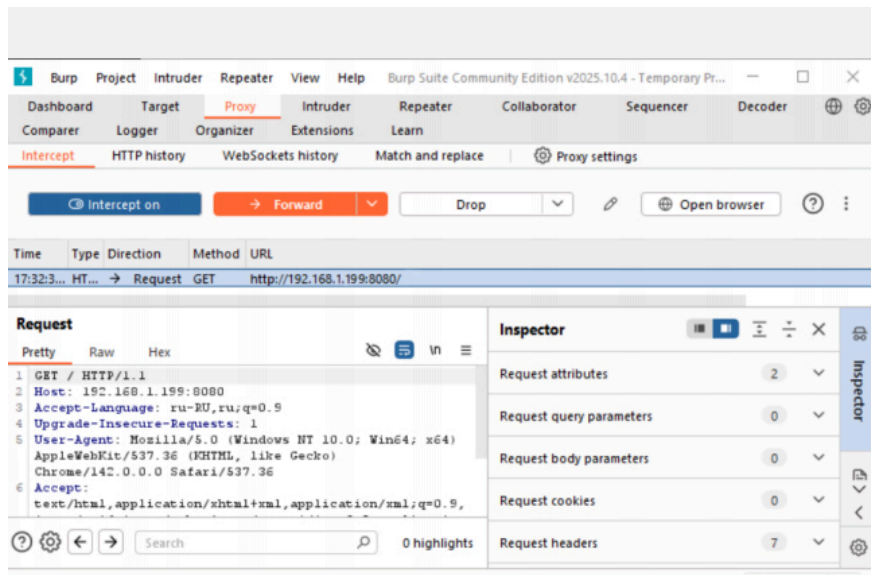


Figure 23. normal HTTP request inside Burp Suite

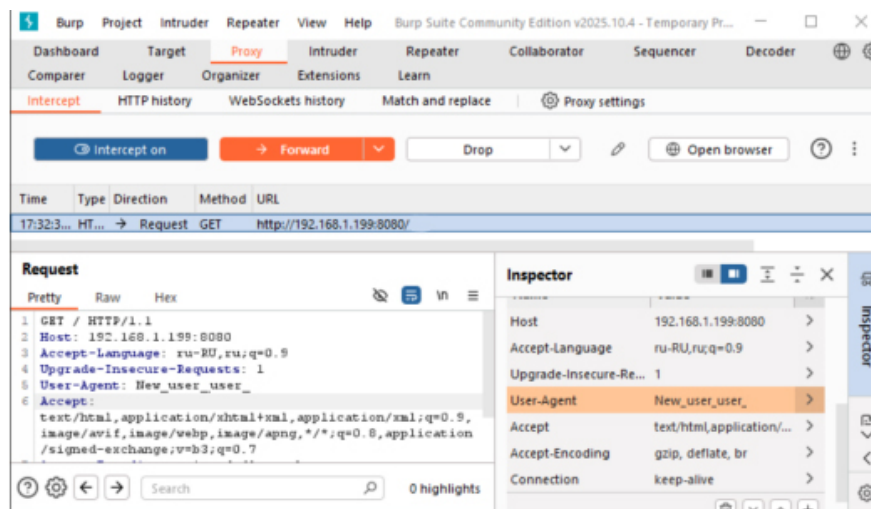


Figure 24. modified HTTP request inside Burp Suite

This request, figure 24, is modifiable because is not encrypted.

Burp Suite sits in the middle as a proxy, it catches that text before it reaches the server, lets me change whatever I want (headers, parameters, even the HTML), and then sends my modified version to the server.

BUT With SSH it's impossible because everything is encrypted end-to-end;

Burp only sees random gibberish and can't touch the real data.

Task 3 - IPv6

1. Configure IPv6 from the Web Server to the Worker. This includes IPs on the servers and the default gateway.

192.168.1.200 - web

192.168.1.199 - admin

192.168.1.198 - worker 1

200 - is the number to change

Configuration on nodes (e.g. web node):

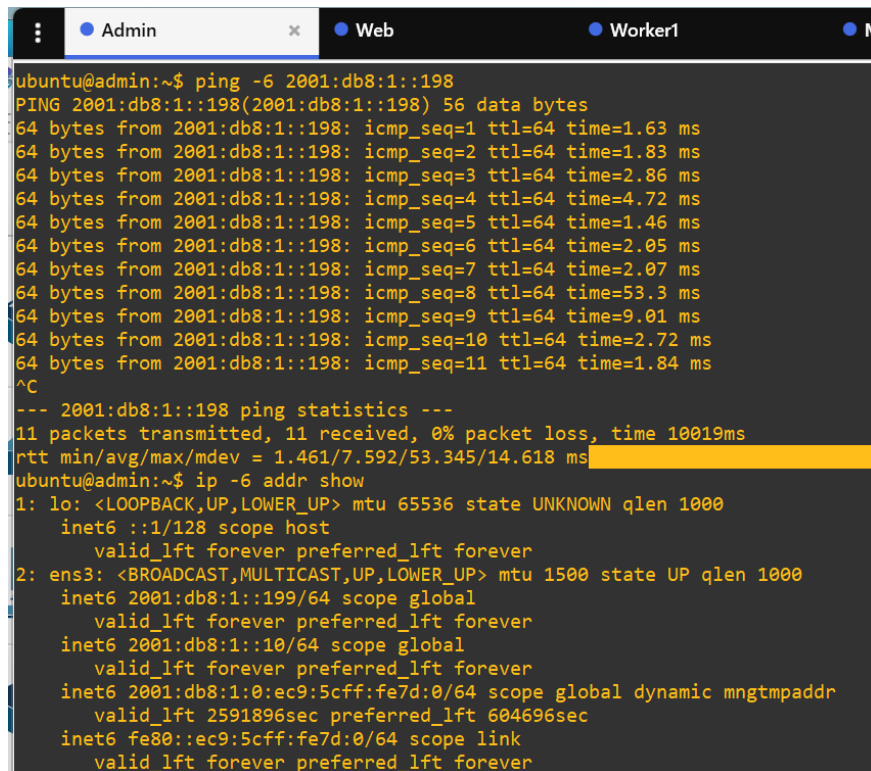
```
sudo ip -6 addr add 2001:db8:1::200/64 dev ens3
sudo ip -6 route add default via 2001:db8:1::1
```

Configuration on the router:

```
/ipv6 address add address=2001:db8:1::1/64 interface=LAN-Bridge advertise=yes
/ipv6 route add dst-address=::/0 gateway=fe80::1%ether3
/ipv6 firewall nat add chain=srcnat action=masquerade out-interface=ether3

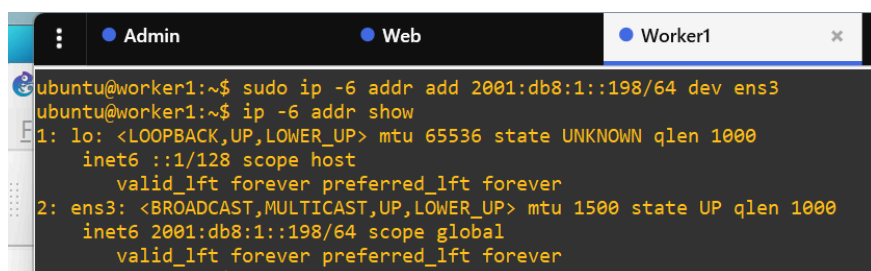
/ipv6 firewall filter
add chain=input action=accept connection-state=established,related
add chain=input action=accept in-interface=LAN-Bridge
add chain=input action=accept protocol=tcp dst-port=80 in-interface=LAN-Bridge
add chain=forward action=accept connection-state=established,related
add chain=forward action=accept in-interface=LAN-Bridge
```

```
add chain=forward action=accept out-interface=LAN-Bridge
add chain=input action=drop
add chain=forward action=drop
```



```
Admin Web Worker1
ubuntu@admin:~$ ping -6 2001:db8:1::198
PING 2001:db8:1::198(2001:db8:1::198) 56 data bytes
64 bytes from 2001:db8:1::198: icmp_seq=1 ttl=64 time=1.63 ms
64 bytes from 2001:db8:1::198: icmp_seq=2 ttl=64 time=1.83 ms
64 bytes from 2001:db8:1::198: icmp_seq=3 ttl=64 time=2.86 ms
64 bytes from 2001:db8:1::198: icmp_seq=4 ttl=64 time=4.72 ms
64 bytes from 2001:db8:1::198: icmp_seq=5 ttl=64 time=1.46 ms
64 bytes from 2001:db8:1::198: icmp_seq=6 ttl=64 time=2.05 ms
64 bytes from 2001:db8:1::198: icmp_seq=7 ttl=64 time=2.07 ms
64 bytes from 2001:db8:1::198: icmp_seq=8 ttl=64 time=53.3 ms
64 bytes from 2001:db8:1::198: icmp_seq=9 ttl=64 time=9.01 ms
64 bytes from 2001:db8:1::198: icmp_seq=10 ttl=64 time=2.72 ms
64 bytes from 2001:db8:1::198: icmp_seq=11 ttl=64 time=1.84 ms
^C
--- 2001:db8:1::198 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10019ms
rtt min/avg/max/mdev = 1.461/7.592/53.345/14.618 ms
ubuntu@admin:~$ ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 2001:db8:1::199/64 scope global
        valid_lft forever preferred_lft forever
    inet6 2001:db8:1::10/64 scope global
        valid_lft forever preferred_lft forever
    inet6 2001:db8:1:0:ec9:5cff:fe7d:0/64 scope global dynamic mngtmpaddr
        valid_lft 2591896sec preferred_lft 604696sec
    inet6 fe80::ec9:5cff:fe7d:0/64 scope link
        valid_lft forever preferred_lft forever
```

Figure 25. Pinging Web node from Admin node, also I show ip6 of Admin machine



```
Admin Web Worker1
ubuntu@worker1:~$ sudo ip -6 addr add 2001:db8:1::198/64 dev ens3
ubuntu@worker1:~$ ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 2001:db8:1::198/64 scope global
        valid_lft forever preferred_lft forever
```

Figure 26. Configuring Web node's ip6 and showing the results.

```

[admin@MikroTik] /ipv6/firewall/filter> /ipv6 route print
Flags: D - DYNAMIC; A - ACTIVE; c - CONNECT, s - STATIC
Columns: DST-ADDRESS, GATEWAY, DISTANCE
#   DST-ADDRESS      GATEWAY      DISTANCE
0   As ::/0          fe80::1%ether3 1
   DAc ::1/128       lo           0
   DAc 2001:db8:1::/64 LAN-Bridge    0
   DAc fe80::%ether3/64 ether3        0
   DAc fe80::%LAN-Bridge/64 LAN-Bridge    0
[admin@MikroTik] /ipv6/firewall/filter> /ipv6 address print
Flags: D - DYNAMIC; G - GLOBAL, L - LINK-LOCAL
Columns: ADDRESS, INTERFACE, ADVERTISE
#   ADDRESS          INTERFACE    ADVERTISE
0   D ::1/128         lo          no
1   DL fe80::e5c:3dff:fe2:0/64 LAN-Bridge  no
2   DL fe80::e5c:3dff:fe2:2/64 ether3      no
3   G 2001:db8:1::/64 LAN-Bridge  yes

```

Figure 27. Configuration on MikroTik

```

ubuntu@ubuntu-cloud:~$ ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
   inet6 2001:db8:1::200/64 scope global dadfailed tentative
       valid_lft forever preferred_lft forever
   inet6 2001:db8:1::e8a:daff:fe64:0/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 2591701sec preferred_lft 604501sec

```

Figure 27. Configuration on Web node.

2. Access the Web's http page using IPv6 from Admin while capturing the traffic again. Can you see the difference? What's the difference in packages? Explain.

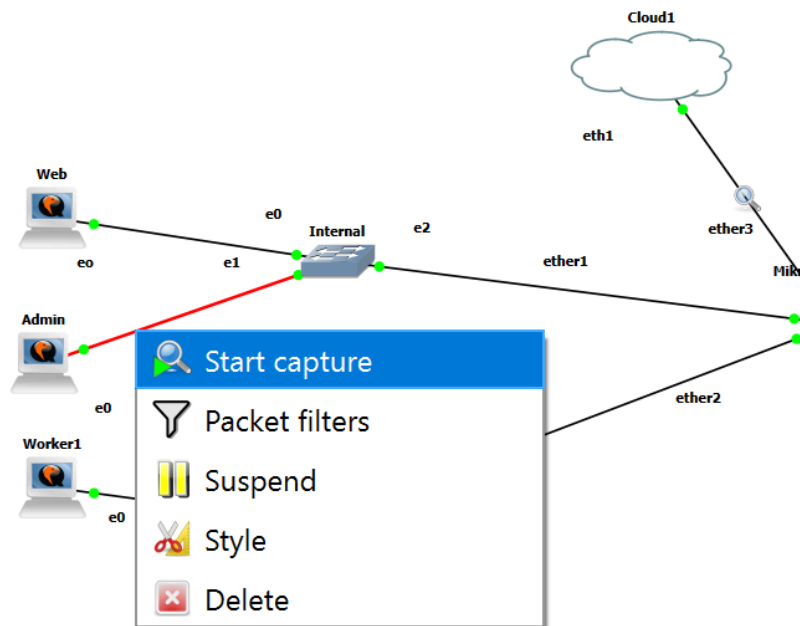


Figure 28. Topology for capturing traffic

```
ubuntu@ubuntu-cloud:~$ sudo sed -i 's/listen 80;/listen [::]:80; listen 80;/' /etc/nginx/sites-enabled/default
^C
ubuntu@ubuntu-cloud:~$ sudo sed -i 's/listen 80;/listen [::]:80; listen 80;/' /etc/nginx/sites-enabled/default
sudo: unable to resolve host ubuntu-cloud: Name or service not known
```

```
ubuntu@ubuntu-cloud:~$ sudo sysctl -w net.ipv6.conf.ens3.accept_dad=0
sudo sysctl -w net.ipv6.conf.ens3.dad_transmits=0
```

```
ubuntu@ubuntu-cloud:~$ ss -tulnp
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 0.0.0.0:34515 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:68 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:68 0.0.0.0:*
udp UNCONN 0 0 192.168.1.200%ens3:68 0.0.0.0:*
tcp LISTEN 0 511 0.0.0.0:80 0.0.0.0:*
tcp LISTEN 0 128 0.0.0.0:22 0.0.0.0:*
tcp LISTEN 0 511 [::]:80 [::]:*
tcp LISTEN 0 128 [::]:22 [::]:*
```

Figure 29. Configuring nginx to listen for ipv6 on Web node

A terminal window with a dark background and light-colored text. The window has a title bar with tabs labeled 'Worker1', 'Admin', and 'Web', and a status bar on the right showing 'MikroTikCHR7.16-1'. The terminal shows a user at the 'ubuntu@admin:~\$' prompt. They run 'ping -6 2001:db8:1::200', which shows three successful ping responses with times around 3.83ms, 1.49ms, and 1.20ms. Then they run 'curl -6 http://[2001:db8:1::200]', which returns an HTML document. The HTML content includes a title 'Welcome to nginx!', a style block with width 35em, and a body with a welcome message, instructions on how to verify the installation, links to nginx.org and nginx.com, and a thank you message.

```
ubuntu@admin:~$ ping -6 2001:db8:1::200
PING 2001:db8:1::200(2001:db8:1::200) 56 data bytes
64 bytes from 2001:db8:1::200: icmp_seq=1 ttl=64 time=3.83 ms
64 bytes from 2001:db8:1::200: icmp_seq=2 ttl=64 time=1.49 ms
64 bytes from 2001:db8:1::200: icmp_seq=3 ttl=64 time=1.20 ms
^C
--- 2001:db8:1::200 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.198/2.171/3.825/1.175 ms
ubuntu@admin:~$ curl -6 http://[2001:db8:1::200]
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
    body {
        width: 35em;
        margin: 0 auto;
        font-family: Tahoma, Verdana, Arial, sans-serif;
    }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
ubuntu@admin:~$
ubuntu@admin:~$
```

Figure 30. Accessing nginx server on Web from Admin

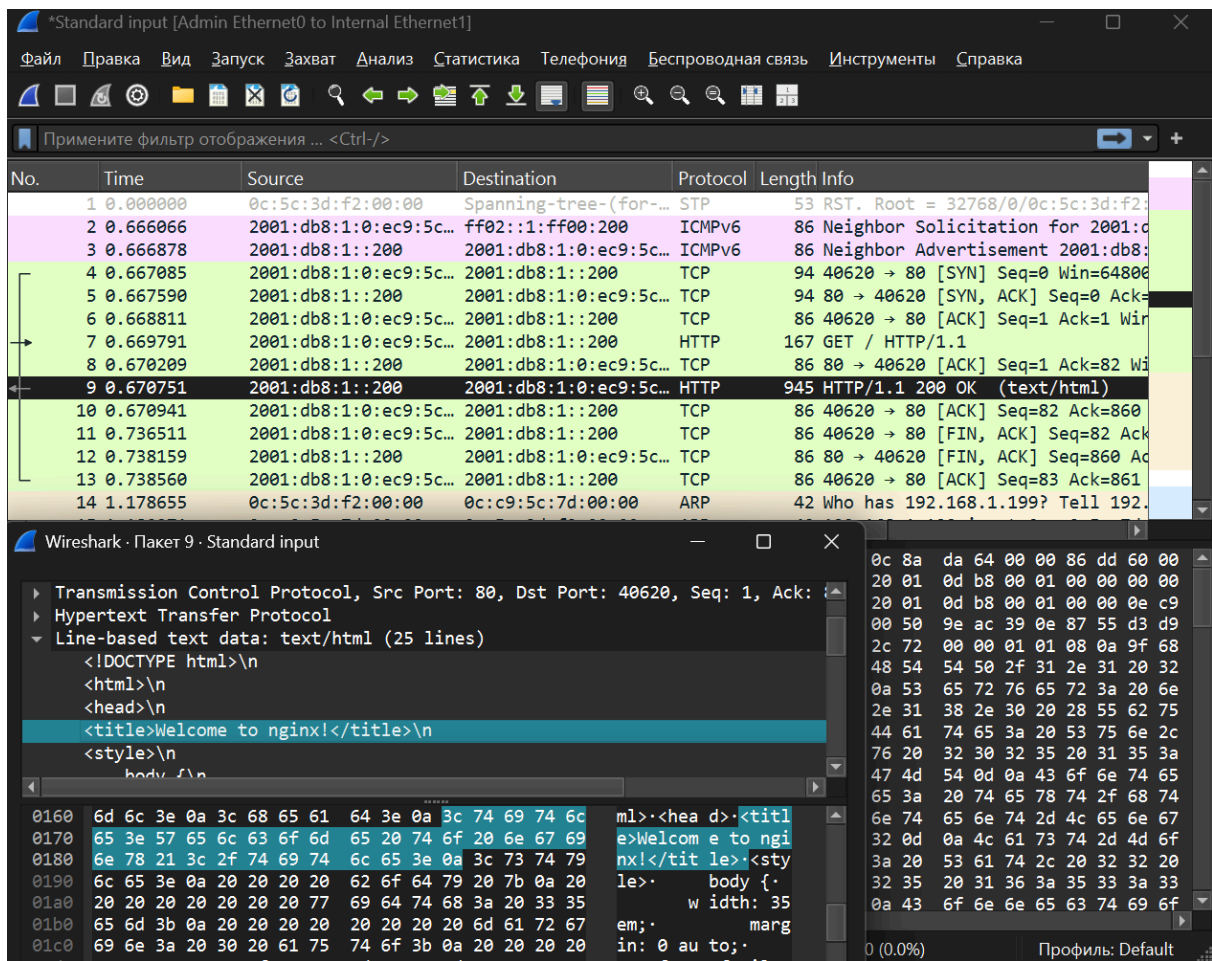


Figure 31. Captured packets HTTP

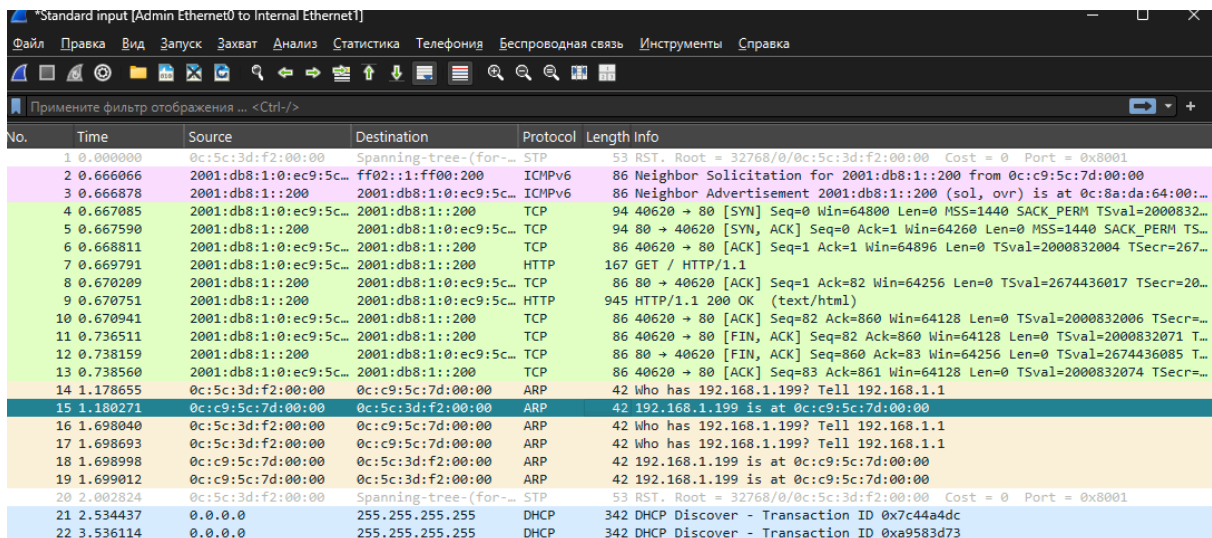


Figure 32. Acknowledgment packets (ICMPv6, ARP with ipv4 addresses, DHCP)

IPv4 and IPv6 differ mainly at the network layer. IPv4 uses a variable-sized header (20–60 bytes), while IPv6 always has a fixed 40-byte header.

The upper layers (TCP, HTTP) stay exactly the same, and instead of **ARP**, IPv6 uses ICMPv6 Neighbor Discovery (NDP) messages to find out which MAC address belongs to an IP.

3. Practice in IPv6 addresses compressing and decompressing. Write your used IPv6 addresses both in full and compact mode. Provide the calculation chain. Attach your IPv6 captures in a folder captures with your report.

Exemplary IPv6 addresses

Web server: 2001:db8:1::100/64

Worker node: 2001:db8:1::200/64

MikroTik LAN: 2001:db8:1::1/64

Compressed: 2001:db8:1::100

Decompression step by step:

1. Compressed form has 4 groups → need 8 groups → replace :: with four zero groups:
2001:db8:1:0000:0000:0000:0000:100
2. I pad every group to 4 digits with leading zeros:
2001:0db8:0001:0000:0000:0000:0000:0100
Full (uncompressed) address of Web server:
2001:0db8:0001:0000:0000:0000:0000:0100
Worker node (2001:db8:1::200)
Compressed: 2001:db8:1::200

Decompression step by step:

1. I replace :: with four zero groups:
2001:db8:1:0000:0000:0000:0000:200
2. Pad to 4 digits per group:
2001:0db8:0001:0000:0000:0000:0000:0200

Full (uncompressed) address of Worker:
2001:0db8:0001:0000:0000:0000:0000:0200
MikroTik LAN gateway (2001:db8:1::1)
Compressed: 2001:db8:1::1
Full: 2001:0db8:0001:0000:0000:0000:0000:0001

Application / References

- [1] Documentation for GNS3 : for installing gns3 on VM :
<https://docs.gns3.com/docs/getting-started/installation/linux/>
- [2] Official site of GNS3 : for installing distribution
- [3] Official site of VMware and their provider : to install VMware Workstation
- [4] GPT for assistance in practical part and explaining theory.

▼ Task 0 setup of topology

To configure topology I also increased system hardware: 2 processors and 4096MB memory.

At figure 0, I am configuring image for a virtual router. The default credentials: Username " **admin** " and an empty password, but I configured " **admin** " (" **ADMIN** ") as password.

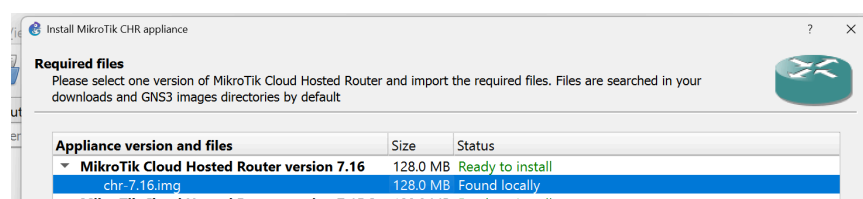


Figure 0. Installing virtual router by MikroTik

Running topology is present at figure 1. The default credentials: Username " `ubuntu` " and " `ubuntu` " password.

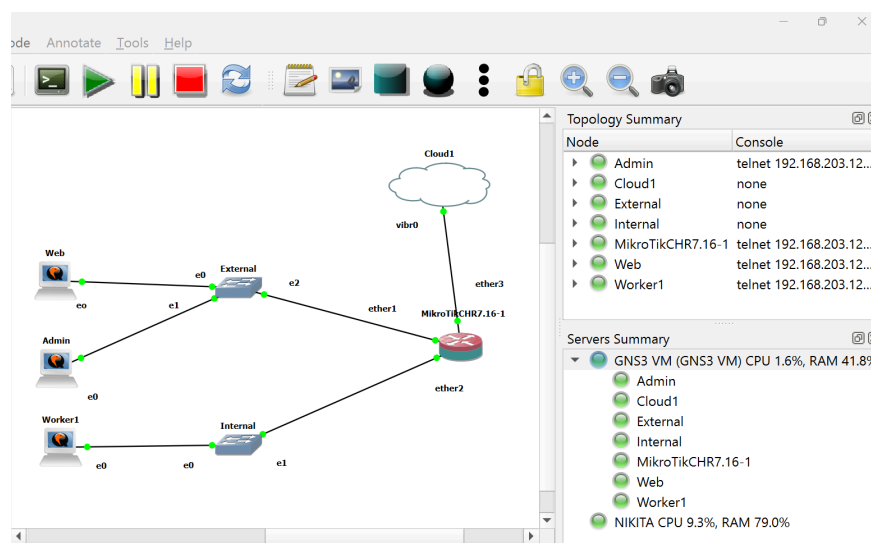


Figure 1. Network topology setup

192.168.122.1 - is for web

```

1: veth: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
   link/ether 52:54:00:49:7d:c5 brd ff:ff:ff:ff:ff:ff
2: veth: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
   link/ether 52:54:00:49:7d:c5 brd ff:ff:ff:ff:ff:ff
3: veth: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
   link/ether 52:54:00:49:7d:c5 brd ff:ff:ff:ff:ff:ff
4: virbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
   link/ether 52:54:00:49:7d:c5 brd ff:ff:ff:ff:ff:ff
   inet 192.168.122.1/24 brd 192.168.122.255 scope global virbr0
       valid_lft forever preferred_lft forever
5: dockero: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default

```

Figure 2. Network topology setup

Now I setup internet access for router and also setup IPs for my nodes inside 192.168.122.0 network:

- 192.168.122.2 - web
- 192.168.122.3 - admin
- 192.168.122.4 - worker 1
- 192.168.122.243/24 - router

Bellow figures show my ip addresses for machines:

```

ubuntu@ubuntu-cloud:~$ sudo netplan apply
WARNING:root:Cannot call Open vSwitch: ovnsdb-server.service is not running
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UP
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 0c:e6:9a:f5:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 192.168.122.2/24 brd 192.168.122.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::e6:9a:f5:00:00/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$

ubuntu@ubuntu-cloud:~$ sudo netplan apply
WARNING:root:Cannot call Open vSwitch: ovnsdb-server.service is not running
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UP
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 0c:e6:9a:f5:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 192.168.122.2/24 brd 192.168.122.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::e6:9a:f5:00:00/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$

ubuntu@ubuntu-cloud:~$ sudo netplan apply
WARNING:root:Cannot call Open vSwitch: ovnsdb-server.service is not running
ubuntu@ubuntu-cloud:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UP
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 0c:e6:9a:f5:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 192.168.122.2/24 brd 192.168.122.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::e6:9a:f5:00:00/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-cloud:~$

```

Figure 3. Network topology setup

eth3 is connects router to cloud, to prove i print interfaces. then I give the router IP of Cloud and check status, figure 4.5.

```

[admin@Mikrotik] > /interface print
Flags: R - RUNNING
Columns: NAME, TYPE, ACTUAL-MTU, MAC-ADDRESS
# NAME TYPE ACTUAL-MTU MAC-ADDRESS
0 R ether1 ether 1500 0C:5C:3D:F2:00:00
1 R ether2 ether 1500 0C:5C:3D:F2:00:01
2 R ether3 ether 1500 0C:5C:3D:F2:00:02
3 ether4 ether 1500 0C:5C:3D:F2:00:03
4 ether5 ether 1500 0C:5C:3D:F2:00:04
5 ether6 ether 1500 0C:5C:3D:F2:00:05
6 ether7 ether 1500 0C:5C:3D:F2:00:06
7 ether8 ether 1500 0C:5C:3D:F2:00:07
8 R lo loopback 65536 00:00:00:00:00:00
[admin@Mikrotik] > /ip dhcp-client add interface=ether3 disabled=no
[admin@Mikrotik] > /ip dhcp-client print
Columns: INTERFACE, USE-PEER-DNS, ADD-DEFAULT-ROUTE, STATUS, ADDRESS
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes searching...
1 ether3 yes yes bound 192.168.122.243/24
[admin@Mikrotik] > /ip dhcp-client print
Columns: INTERFACE, USE-PEER-DNS, ADD-DEFAULT-ROUTE, STATUS, ADDRESS
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes searching...
1 ether3 yes yes bound 192.168.122.243/24
[admin@Mikrotik] >

```

Figure 4. Configuration for virtual router to the internet

```

[admin@Mikrotik] > ip address print
Flags: D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
0 D 192.168.122.243/24 192.168.122.0 ether3
[admin@Mikrotik] > ping 8.8.8.8
  SEQ HOST                      SIZE TTL TIME          STATUS
  ---
    0 8.8.8.8                    56 127 37ms117us
    1 8.8.8.8                    56 127 35ms612us
    2 8.8.8.8                    56 127 35ms689us
    3 8.8.8.8                    56 127 35ms607us
    4 8.8.8.8                    56 127 35ms583us
  sent=5 received=5 packet-loss=0% min-rtt=35ms583us avg-rtt=35ms921us max-rtt=37ms117us
[admin@Mikrotik] > /ip route print
Flags: D - DYNAMIC; A - ACTIVE; c - CONNECT, d - DHCP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
DST-ADDRESS GATEWAY DISTANCE
DAd 0.0.0.0/0 192.168.122.1 1
DAc 192.168.122.0/24 ether3 0

```

Figure 5. Status of connection to network

```
[admin@MikroTik] > /ip firewall filter add chain=input in-interface=ether3 protocol=icmp action=accept
[admin@MikroTik] > /ip firewall filter add chain=input in-interface=ether3 protocol=tcp dst-port=53 action=accept
[admin@MikroTik] > /ip firewall filter add chain=input in-interface=ether3 protocol=udp dst-port=53 action=accept
[admin@MikroTik] > /ip firewall filter add chain=input in-interface=ether3 protocol=tcp dst-port=80,443 action=accept
[admin@MikroTik] > /ip firewall filter add chain=input connection-state=established,related action=accept
[admin@MikroTik] > /ip firewall filter add chain=input connection-state=established,related action=accept
[admin@MikroTik] > /ip firewall filter add chain=input in-interface=ether3 action=drop
```

Figure 6. Setup firewall to allow dns and http(s): router is invisible while allowing outbound traffic

```
[admin@MikroTik] > /ip firewall nat add chain=srcnat out-interface=ether3 action=masquerade
[admin@MikroTik] > /ip firewall nat print
Flags: X - disabled, I - invalid; D - dynamic
0 chain=srcnat action=masquerade out-interface=ether3
```

Figure 7. configuration to translate traffic from lan to wan

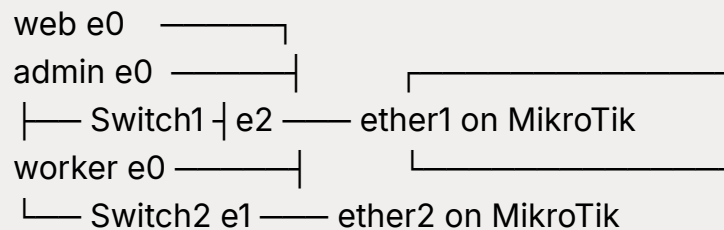
assign a LAN IPs on the router

```
[admin@MikroTik] > /ip address add address=192.168.122.3/24 interface=ether1
[admin@MikroTik] > /ip address add address=192.168.122.2/24 interface=ether1
[admin@MikroTik] > /ip address add address=192.168.122.4/24 interface=ether2
```



my prompt of topology

I use gns3. I have GNS3VM. I have created visually topology, now I need to configure 3 nodes and a router that nodes will have internet access and connectivity. Also my router on mikrotez connects via ether3 to vibr0 of cloud element



▼ Commands to setup mikrotex

```
# Reset configuration if you want to start clean (no defaults)
/system reset-configuration no-defaults=yes

# === Interfaces ===
/interface bridge
add name=LAN-Bridge

# Add both internal ports to the bridge (L2 transparency, like a real switch)
/interface bridge port
add bridge=LAN-Bridge interface=ether1
add bridge=LAN-Bridge interface=ether2

# === IP on the bridge (this will be the gateway for the PCs) ===
/ip address
add address=192.168.1.1/24 interface=LAN-Bridge

# === DHCP Server so the nodes get IP automatically (easiest) ===
/ip pool
add name=dhcp_pool ranges=192.168.1.100-192.168.1.200

/ip dhcp-server
add name=dhcp1 interface=LAN-Bridge address-pool=dhcp_pool lea
```

```

se-time=1d

/ip dhcp-server network
add address=192.168.1.0/24 gateway=192.168.1.1 dns-server=8.8.8.8,
1.1.1.1

# === NAT (masquerade) so the PCs reach Internet ===
/ip firewall nat
add chain=srcnat action=masquerade out-interface=ether3 ipsec-po
licy=out,none

# === Get IP on ether3 automatically from your real network (or set s
tatic) ===
# Most common: DHCP client on ether3
/ip dhcp-client
add interface=ether3 add-default-route=yes

# (If you prefer static)
/ip address add address=10.10.10.50/24 interface=ether3 # example
/ip route add gateway=10.10.10.1

# === DNS (optional, but nice) ===
/ip dns set servers=8.8.8.8,1.1.1.1 allow-remote-requests=yes

```

▼ resolving issue with bad internet connection

mikrotex

```

/interface print
/interface bridge port print
/ip address print
/ip dhcp-server lease print detail
/ip dhcp-server network print
/ip route print
/ip firewall nat print
/ip firewall filter print

```

```
/ip firewall mangle print
/ip dhcp-client print
tool/sniffer/quick interface=ether3
```

nodes

```
sudo cat /etc/netplan/50-cloud-init.yaml
ip link
ip addr
ip route
cat /etc/resolv.conf
ping -c 20 192.168.1.1
ping -c 20 8.8.8.8
traceroute -n 8.8.8.8
cat /etc/netplan/*.yaml 2>/dev/null || cat /etc/cloud/cloud.cfg.d/*.cfg
2>/dev/null || echo "no netplan/cloud-init files"
sudo dmesg | tail -20
cat /etc/netstat/50-
```

▼ current situation on elements

worker

```
sudo cat /etc/netplan/50-cloud-init.yaml

# This file is generated from information provided by the datasource.
# Changes to it will not persist across an instance reboot. To disable
# cloud-init's network configuration capabilities, write a file
# /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg with the
# following:
# network: {config: disabled}
# /etc/netplan/50-cloud-init.yaml
network:
```

```
version: 2
renderer: networkd
ethernets:
  ens3:
    dhcp4: yes
    addresses:
      - 192.168.122.4/24
    routes:
      - to: default
        via: 192.168.122.243
    nameservers:
      addresses: [1.1.1.1, 8.8.8.8]
    match:
      macaddress: 0c:44:cf:68:00:00
    set-name: ens3
```

ubuntu@ubuntu-cloud:~\$

ip link

ip addr

ip route

cat /etc/resolv.conf

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
   UNKNOWN mode DEFAULT group default qlen 1000
```

```
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
```

```
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
```

```
    link/ether 0c:44:cf:68:00:00 brd ff:ff:ff:ff:ff:ff
```

```
    altname enp0s3
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
   UNKNOWN group default qlen 1000
```

```
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
```

```
    inet 127.0.0.1/8 scope host lo
```

```
        valid_lft forever preferred_lft forever
```

```
    inet6 ::1/128 scope host
```

```
        valid_lft forever preferred_lft forever
```

```
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
```

```
    link/ether 0c:44:cf:68:00:00 brd ff:ff:ff:ff:ff:ff
```

```

    altname enp0s3
    inet 192.168.122.4/24 brd 192.168.122.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet 192.168.1.200/24 metric 100 brd 192.168.1.255 scope global
dynamic ens3
    valid_lft 85576sec preferred_lft 85576sec
    inet6 fe80::e44:cfff:fe68:0/64 scope link
        valid_lft forever preferred_lft forever
default via 192.168.122.243 dev ens3 proto static
default via 192.168.1.1 dev ens3 proto dhcp src 192.168.1.200 metri
c 100
1.1.1.1 via 192.168.1.1 dev ens3 proto dhcp src 192.168.1.200 metric 1
00
8.8.8.8 via 192.168.1.1 dev ens3 proto dhcp src 192.168.1.200 metri
c 100
192.168.1.0/24 dev ens3 proto kernel scope link src 192.168.1.200
metric 100
192.168.1.1 dev ens3 proto dhcp scope link src 192.168.1.200 metri
c 100
192.168.122.0/24 dev ens3 proto kernel scope link src 192.168.122.
4
nameserver 127.0.0.53
nameserver 192.168.1.0
nameserver 1.1.1.1
nameserver 8.8.8.8
options edns0 trust-ad
ubuntu@ubuntu-cloud:~$ ping -c 20 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=1.99 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=1.16 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=0.805 ms
^C
--- 192.168.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 0.805/1.318/1.987/0.494 ms
ubuntu@ubuntu-cloud:~$ ping -c 20 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=126 time=93.6 ms

```



```
64 bytes from 8.8.8.8: icmp_seq=2 ttl=126 time=86.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=126 time=124 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=126 time=39.9 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=126 time=50.8 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=126 time=52.0 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=126 time=43.1 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=126 time=48.0 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=126 time=55.1 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=126 time=62.1 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=126 time=80.4 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=126 time=85.7 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=126 time=136 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=126 time=54.6 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=126 time=61.7 ms
^C
--- 8.8.8.8 ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14041ms
rtt min/avg/max/mdev = 39.896/71.587/135.647/28.069 ms
ubuntu@ubuntu-cloud:~$ traceroute -n 8.8.8.8
traceroute: command not found
ubuntu@ubuntu-cloud:~$ sudo traceroute -n 8.8.8.8
sudo: traceroute: command not found
ubuntu@ubuntu-cloud:~$ ping -c 10 google.com
PING forcesafesearch.google.com (216.239.38.120) 56(84) bytes
of data.
From ubuntu-cloud (192.168.122.4) icmp_seq=1 Destination Host Unreachable
From ubuntu-cloud (192.168.122.4) icmp_seq=2 Destination Host Unreachable
From ubuntu-cloud (192.168.122.4) icmp_seq=3 Destination Host Unreachable
From ubuntu-cloud (192.168.122.4) icmp_seq=4 Destination Host Unreachable
From ubuntu-cloud (192.168.122.4) icmp_seq=5 Destination Host Unreachable
From ubuntu-cloud (192.168.122.4) icmp_seq=6 Destination Host Unreachable
```

```
From ubuntu-cloud (192.168.122.4) icmp_seq=7 Destination Host Unreachable
From ubuntu-cloud (192.168.122.4) icmp_seq=8 Destination Host Unreachable
From ubuntu-cloud (192.168.122.4) icmp_seq=9 Destination Host Unreachable
From ubuntu-cloud (192.168.122.4) icmp_seq=10 Destination Host Unreachable
```

```
--- forcesafesearch.google.com ping statistics ---
```

```
10 packets transmitted, 0 received, +10 errors, 100% packet loss,
time 9847ms
```

```
pipe 3          cat /etc/netplan/*.yaml 2>/dev/null || cat /etc/cloud/cloud.cfg.d/*.cfg 2>/dev/null || echo "no"
```

```
## This yaml formatted config file handles setting dev/null || cat /etc/cloud/cloud.cfg.d/*.cfg 2>/dev/null || echo "no"
```

```
## logger information. The values that are necessary to be set
## are seen at the bottom. The top '_log' are only used to remove
## redundancy in a syslog and fallback-to-file case.
```

```
##
```

```
## The 'log_cfgs' entry defines a list of logger configs
```

```
## Each entry in the list is tried, and the first one that
```

```
## works is used. If a log_cfg list entry is an array, it will
```

```
## be joined with '\n'.
```

```
_log:
```

```
- &log_base |
```

```
  [loggers]
```

```
  keys=root,cloudinit
```

```
  [handlers]
```

```
  keys=consoleHandler,cloudLogHandler
```

```
  [formatters]
```

```
  keys=simpleFormatter,arg0Formatter
```

```
  [logger_root]
```

```
  level=DEBUG
```

```
  handlers=consoleHandler,cloudLogHandler
```

```

[logger_cloudinit]
level=DEBUG
qualname=cloudinit
handlers=
propagate=1

[handler_consoleHandler]
class=StreamHandler
level=WARNING
formatter=arg0Formatter
args=(sys.stderr,)

[formatter_arg0Formatter]
format=%(asctime)s - %(filename)s[%(levelname)s]: %(message)s

[formatter_simpleFormatter]
format=[CLOUDINIT] %(filename)s[%(levelname)s]: %(message)s
- &log_file |
[handler_cloudLogHandler]
class=FileHandler
level=DEBUG
formatter=arg0Formatter
args=('/var/log/cloud-init.log', 'a', 'UTF-8')
- &log_syslog |
[handler_cloudLogHandler]
class=handlers.SysLogHandler
level=DEBUG
formatter=simpleFormatter
args=("/dev/log", handlers.SysLogHandler.LOG_USER)

log_cfgs:
# Array entries in this list will be joined into a string
# that defines the configuration.
#
# If you want logs to go to syslog, uncomment the following line.

```

```

# - [ *log_base, *log_syslog ]
#
# The default behavior is to just log to a file.
# This mechanism that does not depend on a system service to o
perate.
- [ *log_base, *log_file ]
# A file path can also be used.
# - /etc/log.conf

# This tells cloud-init to redirect its stdout and stderr to
# 'tee -a /var/log/cloud-init-output.log' so the user can see output
# there without needing to look on the console.
output: {all: '| tee -a /var/log/cloud-init-output.log'}
# to update this file, run dpkg-reconfigure cloud-init
datasource_list: [ NoCloud
ubuntu@ubuntu-cloud:~$ dmesg | tail -20
dmesg: read kernel buffer failed: Operation not permitted
ubuntu@ubuntu-cloud:~$ sudo dmesg | tail -20
[ 26.728526] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[ 26.778188] floppy0: disk absent or changed during operation
[ 26.778192] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 26.802783] Buffer I/O error on dev fd0, logical block 0, async p
age read
[ 26.856560] floppy0: disk absent or changed during operation
[ 26.856566] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[ 26.892170] floppy0: disk absent or changed during operation
[ 26.892173] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 26.931231] Buffer I/O error on dev fd0, logical block 0, async p
age read
[ 26.970592] floppy0: disk absent or changed during operation
[ 26.970596] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[ 27.005716] floppy0: disk absent or changed during operation
[ 27.005720] blk_update_request: I/O error, dev fd0, sector 0 op

```

```
0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 27.040732] Buffer I/O error on dev fd0, logical block 0, async p
age read
[ 27.185786] ISO 9660 Extensions: Microsoft Joliet Level 3
[ 27.212290] ISO 9660 Extensions: RRIP_1991A
[ 35.382537] loop3: detected capacity change from 0 to 8
[ 35.790065] kauditd_printk_skb: 29 callbacks suppressed
[ 35.790067] audit: type=1400 audit(1763725608.092:41): appar
mor="STATUS" operation="profile_replace" profile="uncon"
[ 35.791347] audit: type=1400 audit(1763725608.096:42): appar
mor="STATUS" operation="profile_replace" profile="uncon"
ubuntu@ubuntu-cloud:~$
```

admin

```
Last login: Thu Nov 20 18:49:26 UTC 2025 on ttyS0
ubuntu@ubuntu-cloud:~$ ip link
ip addr
ip route
cat /etc/resolv.conf
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue stat
e UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdi
sc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 0c:c9:5c:7d:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue stat
e UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdi
sc fq_codel state UP group default qlen 1000
    link/ether 0c:c9:5c:7d:00:00 brd ff:ff:ff:ff:ff:ff
```

```

    altname enp0s3
    inet 192.168.122.3/24 brd 192.168.122.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet 192.168.1.198/24 metric 100 brd 192.168.1.255 scope global
dynamic ens3
    valid_lft 85378sec preferred_lft 85378sec
    inet6 fe80::ec9:5cff:fe7d:0/64 scope link
        valid_lft forever preferred_lft forever
default via 192.168.122.243 dev ens3 proto static
default via 192.168.1.1 dev ens3 proto dhcp src 192.168.1.198 metric
100
1.1.1.1 via 192.168.1.1 dev ens3 proto dhcp src 192.168.1.198 metric 1
00
8.8.8.8 via 192.168.1.1 dev ens3 proto dhcp src 192.168.1.198 metric
100
192.168.1.0/24 dev ens3 proto kernel scope link src 192.168.1.198
metric 100
192.168.1.1 dev ens3 proto dhcp scope link src 192.168.1.198 metric
100
192.168.122.0/24 dev ens3 proto kernel scope link src 192.168.122.
3
# This is /run/systemd/resolve/stub-resolv.conf managed by ma
n:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're lookin
g at
# /etc/resolv.conf and seeing this text, you have followed the sym
link.
#
# This is a dynamic resolv.conf file for connecting local clients to
the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS serv
ers
# currently in use.

```

```
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
```

```
nameserver 127.0.0.53
options edns0 trust-ad
search .
```

```
ubuntu@ubuntu-cloud:~$ ping -c 20 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=1.88 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.997 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=1.65 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=0.880 ms
64 bytes from 192.168.1.1: icmp_seq=5 ttl=64 time=0.947 ms
64 bytes from 192.168.1.1: icmp_seq=6 ttl=64 time=0.920 ms
^C
--- 192.168.1.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 0.880/1.212/1.877/0.397 ms
```

```
ubuntu@ubuntu-cloud:~$ ping -c 20 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=126 time=61.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=126 time=212 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=126 time=695 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=126 time=38.6 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=126 time=38.5 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=126 time=44.2 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=126 time=53.2 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=126 time=38.8 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=126 time=40.0 ms
```

```

^C
--- 8.8.8.8 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8014ms
rtt min/avg/max/mdev = 38.518/135.682/694.514/204.479 ms
ubuntu@ubuntu-cloud:~$ ping -c 20 google.com
PING forcesafesearch.google.com (216.239.38.120) 56(84) bytes
of data.
From ubuntu-cloud (192.168.122.3) icmp_seq=1 Destination Host U
nreachable
From ubuntu-cloud (192.168.122.3) icmp_seq=2 Destination Host
Unreachable
From ubuntu-cloud (192.168.122.3) icmp_seq=3 Destination Host
Unreachable
From ubuntu-cloud (192.168.122.3) icmp_seq=4 Destination Host
Unreachable
From ubuntu-cloud (192.168.122.3) icmp_seq=5 Destination Host
Unreachable
From ubuntu-cloud (192.168.122.3) icmp_seq=6 Destination Host
Unreachable
^C
--- forcesafesearch.google.com ping statistics ---
6 packets transmitted, 0 received, +6 errors, 100% packet loss, ti
me 5067ms
pipe 3          cat /etc/netplan/*.yaml 2>/dev/null || cat /etc/clo
ud/cloud.cfg.d/*.cfg 2>/dev/null || echo "no"
## This yaml formatted config file handles settingdev/null || cat /e
tc/cloud/cloud.cfg.d/*.cfg 2>/dev/null || echo "no"
## logger information. The values that are necessary to be set
## are seen at the bottom. The top '_log' are only used to remove
## redundancy in a syslog and fallback-to-file case.
##
## The 'log_cfgs' entry defines a list of logger configs
## Each entry in the list is tried, and the first one that
## works is used. If a log_cfg list entry is an array, it will
## be joined with '\n'.
_log:
- &log_base |
  [loggers]

```



```

keys=root,cloudinit

[handlers]
keys=consoleHandler,cloudLogHandler

[formatters]
keys=simpleFormatter,arg0Formatter

[logger_root]
level=DEBUG
handlers=consoleHandler,cloudLogHandler

[logger_cloudinit]
level=DEBUG
qualname=cloudinit
handlers=
propagate=1

[handler_consoleHandler]
class=StreamHandler
level=WARNING
formatter=arg0Formatter
args=(sys.stderr,)

[formatter_arg0Formatter]
format=%(asctime)s - %(filename)s[%(levelname)s]: %(messag
e)s

[formatter_simpleFormatter]
format=[CLOUDINIT] %(filename)s[%(levelname)s]: %(messag
e)s
- &log_file |
[handler_cloudLogHandler]
class=FileHandler
level=DEBUG
formatter=arg0Formatter
args=('/var/log/cloud-init.log', 'a', 'UTF-8')
- &log_syslog |

```

```
[handler_cloudLogHandler]
class=handlers.SysLogHandler
level=DEBUG
formatter=simpleFormatter
args=("/dev/log", handlers.SysLogHandler.LOG_USER)
```

log_cfgs:

```
# Array entries in this list will be joined into a string
# that defines the configuration.
#
# If you want logs to go to syslog, uncomment the following line.
# - [ *log_base, *log_syslog ]
#
# The default behavior is to just log to a file.
# This mechanism that does not depend on a system service to o
perate.
- [ *log_base, *log_file ]
# A file path can also be used.
# - /etc/log.conf
```

```
# This tells cloud-init to redirect its stdout and stderr to
# 'tee -a /var/log/cloud-init-output.log' so the user can see output
# there without needing to look on the console.
```

```
output: {all: '| tee -a /var/log/cloud-init-output.log'}
```

```
# to update this file, run dpkg-reconfigure cloud-init
```

```
datasource_list: [ NoCloud, ConfigDrive, OpenNebula, DigitalOcea
n, Azure, AltCloud, OVF, MAAS, GCE, OpenStack, CloudSi]
```

```
ubuntu@ubuntu-cloud:~$ sudo dmesg | tail -20
```

```
[ 27.202912] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
```

```
[ 27.233876] floppy0: disk absent or changed during operation
```

```
[ 27.233878] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x0 phys_seg 1 prio class 0
```

```
[ 27.262498] Buffer I/O error on dev fd0, logical block 0, async p
age read
```

```
[ 27.288448] floppy0: disk absent or changed during operation
```

```
[ 27.288453] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
```

```

[ 27.328092] floppy0: disk absent or changed during operation
[ 27.328095] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 27.367984] Buffer I/O error on dev fd0, logical block 0, async p
age read
[ 27.400172] floppy0: disk absent or changed during operation
[ 27.400176] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[ 27.439377] floppy0: disk absent or changed during operation
[ 27.439380] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 27.477194] Buffer I/O error on dev fd0, logical block 0, async p
age read
[ 27.623525] ISO 9660 Extensions: Microsoft Joliet Level 3
[ 27.627319] ISO 9660 Extensions: RRIP_1991A
[ 33.155984] loop3: detected capacity change from 0 to 8
[ 33.865385] kauditd_printk_skb: 29 callbacks suppressed
[ 33.865387] audit: type=1400 audit(1763725606.348:41): appar
mor="STATUS" operation="profile_replace" profile="uncon"
[ 33.867598] audit: type=1400 audit(1763725606.352:42): appar
mor="STATUS" operation="profile_replace" profile="uncon"
ubuntu@ubuntu-cloud:~$ cat /etc/netplan/50-cloud-init.yaml
cat: /etc/netplan/50-cloud-init.yaml: Permission denied
ubuntu@ubuntu-cloud:~$ sudo cat /etc/netplan/50-cloud-init.ya
ml
# This file is generated from information provided by the datasour
ce. Changes
# to it will not persist across an instance reboot. To disable cloud
-init's
# network configuration capabilities, write a file
# /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg with the f
ollowing:
# network: {config: disabled}
# /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  renderer: networkd
  ethernets:

```

```
ens3:
  dhcp4: yes
  addresses:
    - 192.168.122.3/24
  routes:
    - to: default
      via: 192.168.122.243
  nameservers:
    addresses: [1.1.1.1, 8.8.8.8]
  match:
    macaddress: 0c:c9:5c:7d:00:00
  set-name: ens3
ubuntu@ubuntu-cloud:~$ ^C
```

web

```
ubuntu@ubuntu-cloud:~$ sudo cat /etc/netplan/50-cloud-init.yaml
# This file is generated from information provided by the datasource.
# Changes to it will not persist across an instance reboot. To disable cloud-init's
# network configuration capabilities, write a file
# /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg with the following:
# network: {config: disabled}
# /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  renderer: networkd
  ethernets:
    ens3:
      dhcp4: yes
      addresses:
        - 192.168.122.2/24
      routes:
        - to: default
```

```
via: 192.168.122.243
nameservers:
  addresses: [1.1.1.1, 8.8.8.8]
match:
  macaddress: 0c:e6:9a:f5:00:00
set-name: ens3
```

```
ubuntu@ubuntu-cloud:~$ ip link
ip addr
ip route
cat /etc/resolv.conf
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
UP UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 0c:e6:9a:f5:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
UP UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
fq_codel state UP group default qlen 1000
    link/ether 0c:e6:9a:f5:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 192.168.122.2/24 brd 192.168.122.255 scope global ens3
        valid_lft forever preferred_lft forever
    inet 192.168.1.197/24 metric 100 brd 192.168.1.255 scope global
dynamic ens3
    valid_lft 85744sec preferred_lft 85744sec
    inet6 fe80::ee6:9aff:fef5:0/64 scope link
        valid_lft forever preferred_lft forever
    default via 192.168.122.243 dev ens3 proto static
```

```
default via 192.168.1.1 dev ens3 proto dhcp src 192.168.1.197 metric
100
1.1.1.1 via 192.168.1.1 dev ens3 proto dhcp src 192.168.1.197 metric 1
00
8.8.8.8 via 192.168.1.1 dev ens3 proto dhcp src 192.168.1.197 metric
100
192.168.1.0/24 dev ens3 proto kernel scope link src 192.168.1.197
metric 100
192.168.1.1 dev ens3 proto dhcp scope link src 192.168.1.197 metric
100
192.168.122.0/24 dev ens3 proto kernel scope link src 192.168.122.
2
# This is /run/systemd/resolve/stub-resolv.conf managed by ma
n:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're lookin
g at
# /etc/resolv.conf and seeing this text, you have followed the sym
link.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS serv
ers
# currently in use.
#
# Third party programs should typically not access this file directl
y, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.c
onf(5) in a
# different way, replace this symlink by a static file or a different s
ymlink.
#
# See man:systemd-resolved.service(8) for details about the sup
```

```
ported modes of
# operation for /etc/resolv.conf.
```

```
nameserver 127.0.0.53
options edns0 trust-ad
search .
```

```
ubuntu@ubuntu-cloud:~$ ping -c 20 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=1.19 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=1.41 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=1.10 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=1.13 ms
64 bytes from 192.168.1.1: icmp_seq=5 ttl=64 time=1.23 ms
64 bytes from 192.168.1.1: icmp_seq=6 ttl=64 time=0.771 ms
^C
--- 192.168.1.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5009ms
rtt min/avg/max/mdev = 0.771/1.138/1.409/0.192 ms
```

```
ubuntu@ubuntu-cloud:~$ ping -c 20 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=126 time=38.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=126 time=39.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=126 time=38.4 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=126 time=41.2 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=126 time=42.0 ms
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 38.440/39.979/41.983/1.403 ms
ubuntu@ubuntu-cloud:~$ ping google.com
PING forcesafesearch.google.com (216.239.38.120) 56(84) bytes
of data.
From ubuntu-cloud (192.168.122.2) icmp_seq=1 Destination Host U
nreachable
```

```

From ubuntu-cloud (192.168.122.2) icmp_seq=2 Destination Host
Unreachable
From ubuntu-cloud (192.168.122.2) icmp_seq=3 Destination Host
Unreachable
^C
--- forcesafesearch.google.com ping statistics ---
4 packets transmitted, 0 received, +3 errors, 100% packet loss, ti
me 3316ms
pipe 3
ubuntu@ubuntu-cloud:~$ traceroute -n 8.8.8.8
Command 'traceroute' not found, but can be installed with:
sudo apt install traceroute          # version 1:2.1.0-2, or
sudo apt install inetutils-traceroute # version 2:2.2-2ubuntu0.1
ubuntu@ubuntu-cloud:~$ traceroute -n google.com
Command 'traceroute' not found, but can be installed with:
sudo apt install traceroute          # version 1:2.1.0-2, or
sudo apt install inetutils-traceroute # version 2:2.2-2ubuntu0.1
ubuntu@ubuntu-cloud:~$ cat /etc/netplan/*.yaml 2>/dev/null || ca
t /etc/cloud/cloud.cfg.d/*.cfg 2>/dev/null || echo "no netplan/clou
d-init files"
## This yaml formatted config file handles setting
## logger information. The values that are necessary to be set
## are seen at the bottom. The top '_log' are only used to remove
## redundancy in a syslog and fallback-to-file case.
##
## The 'log_cfgs' entry defines a list of logger configs
## Each entry in the list is tried, and the first one that
## works is used. If a log_cfg list entry is an array, it will
## be joined with '\n'.
_log:
- &log_base |
  [loggers]
  keys=root,cloudinit

  [handlers]
  keys=consoleHandler,cloudLogHandler

  [formatters]

```



```

keys=simpleFormatter,arg0Formatter

[logger_root]
level=DEBUG
handlers=consoleHandler,cloudLogHandler

[logger_cloudinit]
level=DEBUG
qualname=cloudinit
handlers=
propagate=1

[handler_consoleHandler]
class=StreamHandler
level=WARNING
formatter=arg0Formatter
args=(sys.stderr,)

[formatter_arg0Formatter]
format=%(asctime)s - %(filename)s[%(levelname)s]: %(message)s

[formatter_simpleFormatter]
format=[CLOUDINIT] %(filename)s[%(levelname)s]: %(message)s
- &log_file |
[handler_cloudLogHandler]
class=FileHandler
level=DEBUG
formatter=arg0Formatter
args=('/var/log/cloud-init.log', 'a', 'UTF-8')
- &log_syslog |
[handler_cloudLogHandler]
class=handlers.SysLogHandler
level=DEBUG
formatter=simpleFormatter
args=("/dev/log", handlers.SysLogHandler.LOG_USER)

```

```

log_cfgs:
# Array entries in this list will be joined into a string
# that defines the configuration.
#
# If you want logs to go to syslog, uncomment the following line.
# - [ *log_base, *log_syslog ]
#
# The default behavior is to just log to a file.
# This mechanism that does not depend on a system service to o
perate.
- [ *log_base, *log_file ]
# A file path can also be used.
# - /etc/log.conf

# This tells cloud-init to redirect its stdout and stderr to
# 'tee -a /var/log/cloud-init-output.log' so the user can see output
# there without needing to look on the console.
output: {all: '| tee -a /var/log/cloud-init-output.log'}
# to update this file, run dpkg-reconfigure cloud-init
datasource_list: [ NoCloud, ConfigDrive, OpenNebula, DigitalOcea
n, Azure, AltCloud, OVF, MAAS, GCE, OpenStack, CloudSigma, S
martOS, Bigstep, Scaleway, AliYun, Ec2, CloudStack, Hetzner, IB
MCloud, Oracle, Exoscale, RbxCloud, UpCloud, VMware, Vultr, LX
D, NWCS, Akamai, WSL, None ]
ubuntu@ubuntu-cloud:~$
dmesg | tail -20
dmesg: read kernel buffer failed: Operation not permitted
ubuntu@ubuntu-cloud:~$ dmesg | tail -20
dmesg: read kernel buffer failed: Operation not permitted
ubuntu@ubuntu-cloud:~$ sudo dmesg | tail -20
[ 26.905175] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[ 26.934204] floppy0: disk absent or changed during operation
[ 26.934207] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 26.971136] Buffer I/O error on dev fd0, logical block 0, async p
age read
[ 26.999811] floppy0: disk absent or changed during operation

```

```
[ 26.999814] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[ 27.027438] floppy0: disk absent or changed during operation
[ 27.027440] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 27.058317] Buffer I/O error on dev fd0, logical block 0, async p
age read
[ 27.087803] floppy0: disk absent or changed during operation
[ 27.087806] blk_update_request: I/O error, dev fd0, sector 0 op
0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[ 27.125117] floppy0: disk absent or changed during operation
[ 27.125121] blk_update_request: I/O error, dev fd0, sector 0 op 0
x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 27.186947] Buffer I/O error on dev fd0, logical block 0, async p
age read
[ 27.373535] ISO 9660 Extensions: Microsoft Joliet Level 3
[ 27.376502] ISO 9660 Extensions: RRIP_1991A
[ 33.613016] loop3: detected capacity change from 0 to 8
[ 34.119526] kauditd_printk_skb: 29 callbacks suppressed
[ 34.119528] audit: type=1400 audit(1763725606.840:41): appar
mor="STATUS" operation="profile_replace" profile="unconfined"
name="/usr/lib/snapd/snap-confine" pid=665 comm="apparmor_
parser"
[ 34.120974] audit: type=1400 audit(1763725606.840:42): appar
mor="STATUS" operation="profile_replace" profile="unconfined"
name="/usr/lib/snapd/snap-confine//mount-namespace-capture-
helper" pid=665 comm="apparmor_parser"
ubuntu@ubuntu-cloud:~$
```

mikro

```
[admin@MikroTik] > /interface print
Flags: R - RUNNING; S - SLAVE
Columns: NAME, TYPE, ACTUAL-MTU, L2MTU, MAC-ADDRESS
```

```

# NAME      TYPE      ACTUAL-MTU L2MTU MAC-ADDRESS
0 RS ether1  ether      1500      0C:5C:3D:F2:00:00
1 RS ether2  ether      1500      0C:5C:3D:F2:00:01
2 R  ether3  ether      1500      0C:5C:3D:F2:00:02
3  ether4    ether      1500      0C:5C:3D:F2:00:03
4  ether5    ether      1500      0C:5C:3D:F2:00:04
5  ether6    ether      1500      0C:5C:3D:F2:00:05
6  ether7    ether      1500      0C:5C:3D:F2:00:06
7  ether8    ether      1500      0C:5C:3D:F2:00:07
8 R LAN-Bridge bridge    1500 65535 0C:5C:3D:F2:00:00
9 R lo      loopback  65536 00:00:00:00:00:00

[admin@MikroTik] > /interface bridge port print
Columns: INTERFACE, BRIDGE, HW, PVID, PRIORITY, HORIZON
# INTERFACE BRIDGE  HW  PVID PRIORITY HORIZON
0 ether1    LAN-Bridge yes   1 0x80  none
1 ether2    LAN-Bridge yes   1 0x80  none

[admin@MikroTik] > /ip address print
Flags: D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS      NETWORK      INTERFACE
0 192.168.1.1/24 192.168.1.0 LAN-Bridge
1 D 192.168.122.243/24 192.168.122.0 ether3

[admin@MikroTik] > /ip dhcp-server lease print detail
Flags: X - disabled, R - radius, D - dynamic, B - blocked
0 D address=192.168.1.199 address-lists="" server=dhcp1 dhcp-option="" status=conflict expires-after=9h46m43s
    last-seen=14h13m17s active-address=192.168.1.199 active-server=dhcp1 src-mac-address=0C:E6:9A:F5:00:00

1 D address=192.168.1.198 mac-address=0C:C9:5C:7D:00:00 address-lists="" server=dhcp1 dhcp-option="" status=bound
    expires-after=23h59m24s last-seen=36s active-address=192.168.1.198 active-mac-address=0C:C9:5C:7D:00:00
    active-server=dhcp1 host-name="ubuntu-cloud"

2 D address=192.168.1.200 mac-address=0C:44:CF:68:00:00 address-lists="" server=dhcp1 dhcp-option="" status=bound
    expires-after=23h59m24s last-seen=36s active-address=192.1

```

```
68.1.200 active-mac-address=0C:44:CF:68:00:00
active-server=dhcp1 host-name="ubuntu-cloud"
```

```
3 D address=192.168.1.197 mac-address=0C:E6:9A:F5:00:00 address-lists="" server=dhcp1 dhcp-option="" status=bound
expires-after=23h59m25s last-seen=35s active-address=192.168.1.197 active-mac-address=0C:E6:9A:F5:00:00
active-server=dhcp1 host-name="ubuntu-cloud"
```

```
[admin@MikroTik] > /ip dhcp-server network print
```

```
Columns: ADDRESS, GATEWAY, DNS-SERVER
```

```
# ADDRESS      GATEWAY      DNS-SERVER
```

```
0 192.168.1.0/24 192.168.1.1 8.8.8.8
```

```
1.1.1.1
```

```
[admin@MikroTik] > /ip route print
```

```
Flags: D - DYNAMIC; A - ACTIVE; c - CONNECT, d - DHCP
```

```
Columns: DST-ADDRESS, GATEWAY, DISTANCE
```

```
DST-ADDRESS    GATEWAY      DISTANCE
```

```
DAd 0.0.0.0/0   192.168.122.1 1
```

```
DAC 192.168.1.0/24 LAN-Bridge    0
```

```
DAC 192.168.122.0/24 ether3      0
```

```
[admin@MikroTik] > /ip firewall nat print
```

```
Flags: X - disabled, I - invalid; D - dynamic
```

```
0 X chain=srcnat action=masquerade out-interface=ether3 ipsec-policy=out,none
```

```
1 chain=srcnat action=masquerade out-interface=ether3
```

```
2 chain=srcnat action=masquerade src-address=192.168.1.0/24 out-interface=ether3
```

```
3 chain=srcnat action=masquerade src-address=192.168.1.0/24 out-interface=ether3
```

```
4 chain=srcnat action=masquerade src-address=192.168.1.0/24 out-interface=ether3
```

```
5 chain=srcnat action=masquerade src-address=192.168.1.0/24 out-interface=ether3
```

```
6 chain=srcnat action=masquerade src-address=192.168.1.0/24
out-interface=ether3
```

```
[admin@MikroTik] > /ip firewall filter print
```

```
Flags: X - disabled, I - invalid; D - dynamic
```

```
0 chain=input action=accept connection-state=established,related
```

```
1 chain=input action=accept in-interface=LAN-Bridge
```

```
2 chain=forward action=accept connection-state=established,related
```

```
3 chain=forward action=accept in-interface=LAN-Bridge
```

```
4 chain=forward action=drop
```

```
5 chain=input action=drop
```

```
6 chain=forward action=accept connection-state=established,related
```

```
7 chain=forward action=accept in-interface=LAN-Bridge
```

```
8 chain=forward action=drop
```

```
[admin@MikroTik] > /ip firewall mangle print
```

```
Flags: X - disabled, I - invalid; D - dynamic
```

```
[admin@MikroTik] > /ip dhcp-client print
```

```
Flags: I - INVALID
```

```
Columns: INTERFACE, USE-PEER-DNS, ADD-DEFAULT-ROUTE, STATUS, ADDRESS
```

```
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
```

```
;;; DHCP client can not run on slave or passthrough interface!
```

```
0 | ether1 yes yes searching...
```

```
1 | ether3 yes yes bound 192.168.122.243/24
```

```
4
```

```
# because this command is not working /tool torch ether3 rx tx

[admin@MikroTik] > /tool/sniffer/quick interface=ether3
Columns: INTERFACE, TIME, NUM, DIR, SRC-MAC, DST-MAC, SRC-ADDRESS
INTERFACE TIME NUM DIR SRC-MAC DST-MAC
SRC-ADDRESS
ether3 0.449 1 → 0C:5C:3D:F2:00:02 33:33:00:00:00:01 fe
80::e5c:3dff:fef2:2:5678 (discovery)
ether3 0.449 2 → 0C:5C:3D:F2:00:02 FF:FF:FF:FF:FF:FF 1
92.168.122.243:5678 (discovery)
ether3 0.449 3 → 0C:5C:3D:F2:00:02 01:00:0C:CC:CC:CC
ether3 0.449 4 → 0C:5C:3D:F2:00:02 01:80:C2:00:00:0E
```

solution option 1

```
sudo systemctl disable --now systemd-resolved 2>/dev/null
sudo rm -f /etc/resolv.conf
echo "nameserver 192.168.1.1" | sudo tee /etc/resolv.conf
sudo chmod +i /etc/resolv.conf

sudo ip addr flush dev ens3
sudo dhclient -v ens3

sudo touch /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg
echo "network: {config: disabled}" | sudo tee /etc/cloud/cloud.cfg.d/
99-disable-network-config.cfg

sudo rm -f /etc/netplan/50-cloud-init.yaml
sudo netplan apply 2>/dev/null || true
```

solution to solve flush error (change name on each node)

```
# Find the current hostname
hostname
# → probably returns "ubuntu-cloud"

# Edit the hostname file
sudo nano /etc/hostname
# change it to whatever you want, e.g. "web" (for your web node)
# save & exit (Ctrl+O → Enter → Ctrl+X)

# Add the hostname to /etc/hosts
sudo nano /etc/hosts
# add this line (replace "web" with whatever you chose above)
127.0.1.1  web
# save & exit

# Reboot or just apply immediately
sudo hostnamectl set-hostname web
```

fix on gns3vm code

```
sudo apt install -y uml-utilities bridge-utils

# Create persistent TAP user (gns3 will own it)
sudo useradd --system --shell /usr/sbin/nologin tapuser 2>/dev/null || true

# Create the permanent TAP + bridge script
sudo tee /usr/local/bin/gns3-tap-bridge.sh > /dev/null <<'EOF'
#!/bin/bash
# Remove old stuff if exists
ip link del br0 2>/dev/null || true
ip tuntap del tap0 mode tap 2>/dev/null || true
```



```

# Create TAP owned by current user (gns3 runs as your user)
ip tuntap add tap0 mode tap user $(whoami)
ip link set tap0 up promisc on

# Create bridge and add your real NIC + tap0
ip link add br0 type bridge
ip link set tap0 master br0
ip link set eth0 master br0 # ← change to eth0 if your bridged NIC is eth0
ip link set eth0 up
ip link set tap0 up
ip link set br0 up

# Move the IP from eth0 to br0 (keeps Internet on GNS3 VM itself)
ETH_IP=$(ip -4 addr show eth0 | grep -oP '(?<=inet\s)\d+(\.\d+){3}/\d+')
[ -n "$ETH_IP" ] && ip addr del $ETH_IP dev eth0 2>/dev/null
[ -n "$ETH_IP" ] && ip addr add $ETH_IP dev br0 2>/dev/null

# Restore default route via bridge
ip route del default 2>/dev/null || true
GATEWAY=$(ip route | grep default | awk '{print $3}')
[ -n "$GATEWAY" ] && ip route add default via $GATEWAY dev br0
EOF

# Make it executable
sudo chmod +x /usr/local/bin/gns3-tap-bridge.sh

# Make it start at boot (systemd service)
sudo tee /etc/systemd/system/gns3-tap-bridge.service > /dev/null <<'EOF'
[Unit]
Description=GNS3 TAP + Bridge for full Internet speed
After=network-online.target

[Service]
Type=oneshot
ExecStart=/usr/local/bin/gns3-tap-bridge.sh

```

```
RemainAfterExit=yes
```

```
User=root
```

```
[Install]
```

```
WantedBy=multi-user.target
```

```
EOF
```

```
# Enable and start
```

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable --now gns3-tap-bridge.service
```